Internship report

Revealing the hidden correspondence between Queen Marie-Antoinette II and the Count of Fersen : What can deep-learning bring ?





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Introduction

Problematic : looking for a hidden truth

Between end of June 1791 and August 1792, the royal family in France is placed in a supervised residence. In Fact since 1789, France has begun an important change about his political carried by the revolution. During this period, the economic, social and political context is in poor condition : the People and more specifically the Third-Estate suffer about taxes, faminy and inequality. With this difficulties, People gradually lose faith in monarchy and they decide to take the power. With this putsch, the people obtain the power but they don't want to get rid of the king and they decide to establish a constitutional monarchy : an strange period start where the power is sharing between an constitutionnal assembly and the king who can apply veto. The king and the royal family seem to accept the new policy, but in fact they are looking for a way to regain power. The 20 June 1791, The royal family helped by the Earl Fersen, try to flee away to rejoin an army that will help him to reclaim the power. The escape fail, the royal family lose all his powers and they are assigned in a supervised residence.

During this surveillance, Queen Marie-Antoinette II and Earl Fersen will exchange many letters. To protect the exchange, many lines of this letters have been censored by the authors. We estimate that these lines contain some informations that can help us to understand the historic context as the expression of feelings about the revolution or about the imprisonment of the royal family and also about the specific relation between the Queen and the Earl. All of this feelings are very important to understand the context, they are historic testimony and they can reveal us more about political and social tensions of this period.

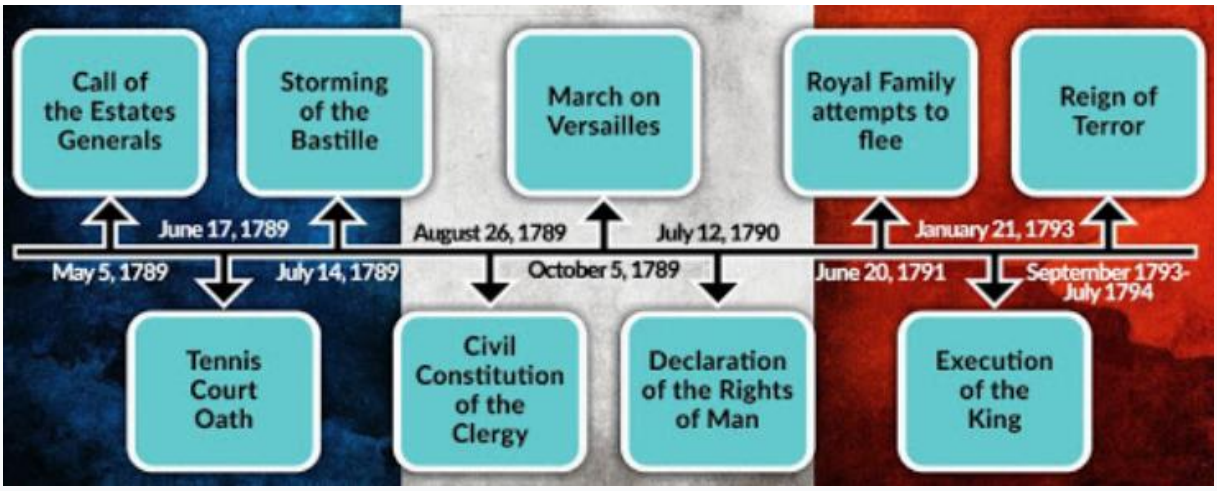


Figure 1 : French revolution Timeline

To find the hidden truth, in 2014, the Foundation of Heritage Sciences (FSP) financed an ambitious project, its realization seemed complex and uncertain. The goal of this project is to extract and exploit the underlying writing of ancient manuscripts. The constraint imposed with this ambitious project is to protect the integrity of the manuscripts. The old manuscripts belong to the cultural

heritage of the country, the treatments applied for the extraction must therefore not damage these documents : these documents are the letters exchanged between the queen and the Earl Fersen.

To reach its goals, the FSP finances the REX I project and manages to read an censored passage from a manuscript : Letter from Marie-Antoinette to the count of Fersen of January 4, 1792, 440AP / 1. With the results of the project REX I, the National Archives, in collaboration with the Center for Research on Conservation (CRC, USR3224, CNRS / MNHN / Ministry of Culture) and the Heritage and Cultural Dynamics Laboratory (DYPAC) decide to continue the study of censored passages of the correspondence between Queen Marie-Antoinette and the Earl of Fersen. The new REX II project (Research on the extraction and use of underlying lines in old manuscripts : application to Marie-Antoinette's letters) aims to reveal the content of the original text of all the censored passages. Without affecting the integrity of the originals.

The acquisition of an XRF scanner (X-Ray Fluorescence scanner, confers *annexe I*), allow to establish the spectral profile of the letters and therefore the maps of the elements present in the inks by X-ray fluorescence spectroscopy. The ink used for writing and censorring the letters are the famous Ferro-Gallic ink : this ink is made up of metallurgical elements. The spectral analysis of the letters makes it possible to highlight the metallurgical composition of the letters.

Using these spectral profiles, the CNRS has already succeeded in extracting new underlying traces allowing to reveal other passages that were previously unreadable. However, due to the small difference in compositions between the inks of the original text and the censored text, the low concentration of the mapped elements, the visualization of the data from the front and the back on the same data, the mappings obtained (for an element or by combining the data by PCA) do not always allow good readability of the original text.

The objective of my internship is to establish a proof of concept allowing to show the contribution of modern machine learning techniques in the detection of the original text in the images resulting from the digitization of letters.

As an intern in the XLIM laboratory of the University of Poitiers, my job consist in apply modern Machine Learning techniques (supervised or not) in order to separate the different sources (paper, censorring ink, writing ink). I have to test, compare and prove the value of using these different methods.

My work follow the REX I and REX II projects. These projects enabled to read a number of censored passages by using various methods. The first project had allowed the reading of a censored passage of a letter using X-ray fluorescence spectroscopy : For this first project the mapping of the copper present only in the ink of the original text was sufficient to reveal the underlying text. However, a number of problems were highlighted during the first project :

- Letters with very similar composition between writing and censorring inks.
- Penetration of X-rays used in XRF which makes it difficult to read the data of censored letters written on both sides.

The second project made it possible to overcome these problems by using different approaches in data processing (combination of data from different elements by statistical tools, morpho-mathematical tools from the field of image processing). My job is to complement this second project, find new solutions from Machine Learning for the problems highlighted by the first project.

The manuscripts studied are a set of correspondence between Queen Marie-Antoinette II and the Count of Fersen, written between the end of June 1791 and August 1792. The majority of this corpus is kept in the national archives : 25 letters from Marie-Antoinette (4 originals autographs, the others being contemporary copies made by Fersen or his secretary) of which 7 have censored passages (representing a total of 53 unreadable lines) and 29 letters from the Earl of Fersen of which 8 are censored (55 lines). XRF analysis was performed on all of the censored passages.

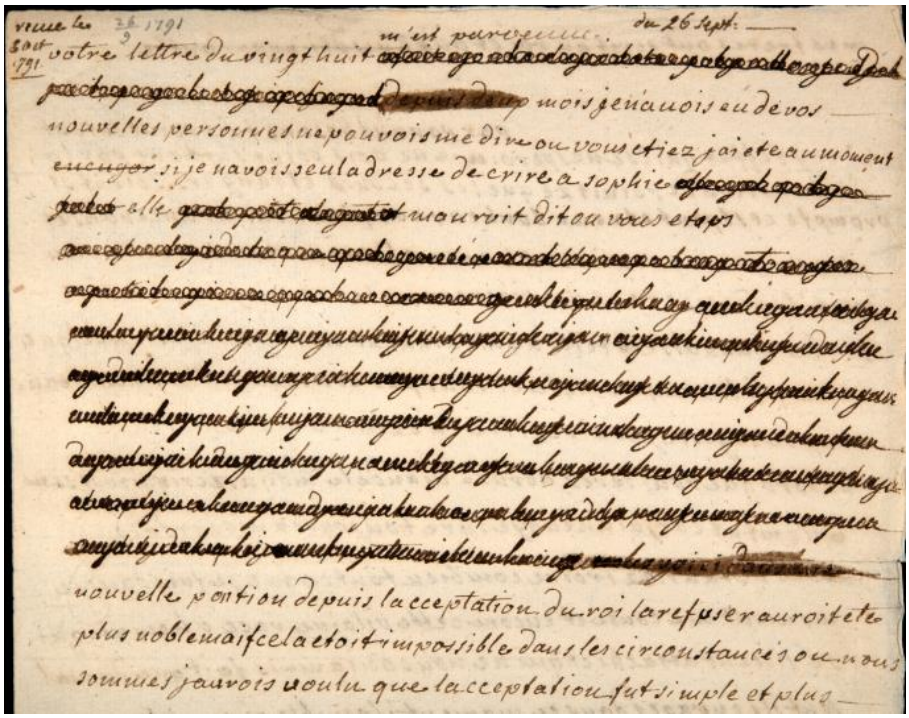


Figure 2 : example of censorring letter (440AP, letter 1_1_8 page 1)

This example is considered as an trivial example because here are a strong difference between the two ink : the copper concentration.

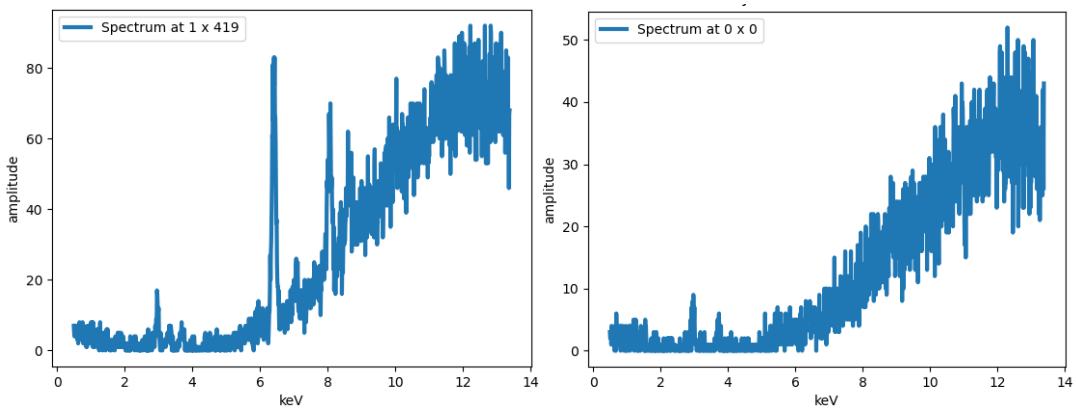


Figure 3 : examples of XRF spectrum

We can see an examples of the XRF spectrum obtained with the XRF analysis. This spectrum will be our main data to separate the different ink.

Contributors and host structure

This project has started in 2014, since this date, many contributors have participated to this. The first one is the Foundation of Heritage Sciences (FSP) that have financed the project in 2014. FSP is a foundation supervised by the Ministry of Culture, its main goal and ambition is to structure and finance research around cultural heritage, targeting three subgoals :

- Increase and research heritage's knowledges
- Upgrade conservation and restoration processes
- Improve knowledge dissemination

FSP is engaged to search and share heritage's knowledges. Following this way, they have financed many research, scientific event or thesis.

The second contributor is the National archives. The missions of the National Archives are the collection, classification, conservation, restoration, communication and enhancement of State archives. This service with national competence depends on the ministry in charge of cultural affairs. since several years, the national archives are part of the scientific landscape through partnerships fostered with various higher education establishments such as Comue Paris-Lumière, Labex Arts-H2H and Hastec or the Fondation des sciences du Patrimoine (FSP) of which they are an associate member. National archives has already contributed, with the FSP, to the Verspera, Rex and poilus Testaments projects.

DYPAC laboratory (Heritage and cultural dynamics) is a laboratory at the University of Versailles-Saint-Quentin : it is specializes in the study of culture and heritage, from Antiquity to the end of the modern era. The DYPAC laboratory carries out scientific projects in collaboration with major French heritage institutions (National Archives of France, National Library of France, Louvre Museum), as well as with experimental science laboratories working in the field of materials expertise. Its research program is structured around the following axes :

- The making of heritage: processes, forms and objects
- Tangible heritage, intangible heritage
- Cultures, heritage and knowledge in Europe

Currently it is working on the digitization of manuscripts which were sometimes badly damaged by the fire of 1944. DYPAC is another contributor to the REX and REX II project.

The CRC laboratory (Center for Research on Preservation) is a research and services unit that falls under the joint umbrella of the CNRS (France's national scientific research council), the French National

Museum of Natural History and the Ministry of Culture and Communication. It specializes in research aiming at the preservation of cultural property. The unit comprises some fifty scientists : chemists, physicists, biologists, art historians and science and technology historians. among these scientists, we find Anne MiCHELIN, lecturer and chemist at "color and visual effects" pole and Christine ANDRAUD, professor and physicist at the same pole. This pole focus the optic measure and modelization to have a better knowledge about composition and structure of heritage's materials.

The last one is the XLIM laboratory, it is also the host structure. This laboratory focused on electronics and microwaves, optics and photonics, mathematics, computer science and images, CAD, in the fields of space, telecom networks, secure environments, bioengineering, new materials, energy and imagery. This laboratory is constituted to 3 scientific pole :

- Electronic
- Fiber photonics and coherent sources
- Mathematics, Computer Science, Image

these poles bring together 6 research axes :

- RF systems
- RF-ELITE
- Intelligent Systems and Grids
- Fiber photonics and coherent sources
- Image synthesis and analysis
- Mathematics & Information Security

This laboratory brings together at round 440 people : intern, doctoral student, Professor, lecturer, searcher. Professor and ICONES Team Manager Philippe CARRE is one of the scientist that work at XLIM laboratory.

stakes of the problematic

The stakes of this problem are multiple. Indeed, from a historical point of view, the underlying tracings of our corpus constitute an additional historical testimony. Also, the methodology used to 4 recover these underlying plots could inspire many people and allow the reading of other redacted manuscripts. Another issue related to this problem is to succeed in extracting information from an old manuscript without altering its physical integrity : old manuscripts and other media do not easily undergo different treatments, but these documents are important because they constitute historical evidence and a cultural heritage, so care must be taken not to damage them. From a scientific point of view, this problem is a challenge to be taken up, the methodology remains to be found and the subject is multidisciplinary.

organization and strategy

Focus on data

To design strategies, we have to know the main characteristics of our datasets. We are talking about a set of datasets: majority of corpus letters are written with different inks. inks metallurgical composition from letter to letter is variable. We do not have any pre-labeled datasets. XRF spectroscopy was performed on all of the censored lines, so there is no spectroscopy on uncensored data. We have tif images of each corpus letter. We have chemical mappings for a set of elements for each censored line : that is to say, we have a black and white image for each characteristic element (sulfur, copper, iron, potassium, manganese, etc.) for each censored line (this information is extracted from XRF spectroscopy, so it is not an additional data but a visualization). Our corpus is of varying complexity : there are trivial cases where the study of an element allows us to separate the sources and there are more complex cases. The influence of the reverse side on the recto side, the use of metallurgical flake to dry the ink, and the composition of the writing and censorring inks are parameters that influence the complexity for source separation.

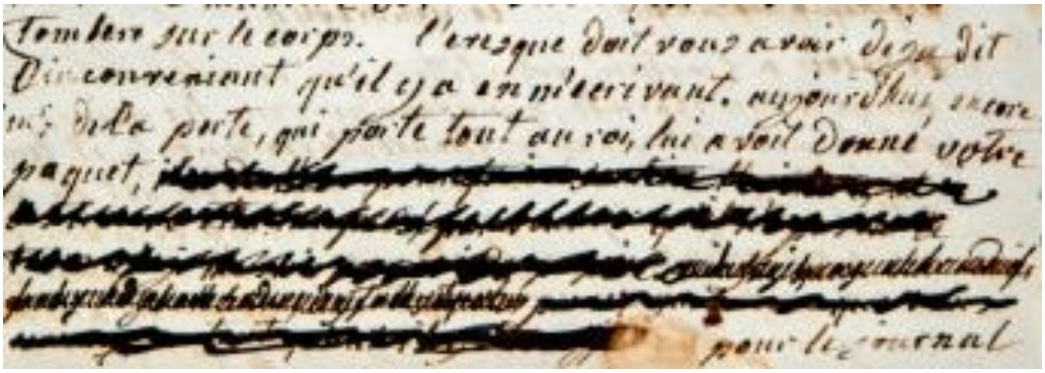


Figure 4 : complex letter with similar written and censorring ink (440AP, letter 1_1_2 page 6)

The letters are written with Ferro-Gallic inks, the main constituents of which are :

- Vegetable tannin (most often based on gall nuts)
- Iron sulphate (sometimes copper sulphate)
- A binder : gum arabic
- A solvent : water
- Additives : coloring agents, alum, wine, vinegar, etc.

The presence of impurity in basic constituents such as the presence of zinc sulphate or manganese with iron sulphate, is one of the factors that explains the differences between inks in our corpus.

Ferro-Gallic inks can degrade manuscripts. In fact, the inks acidity due to the formation of hydronium ions during the ink preparation or the excessive proportion of iron sulphate contributes to manuscripts degradation : browning and deterioration of paper. In addition, Ferro-Gallic ink tends to lose color over time due to the oxidation of its major compounds.

Field expertise tells us that the study of certain chemical elements (corresponding to specific bands of our spectra) and the study of certain specific ratios is sufficiently decisive for our problem. The bands of our spectrum corresponding to the chemical éléments : sulfur (S), potassium (K), manganese (Mn), copper (Cu), iron (Fe) and zinc (Zn) are decisive for our study. These bands will constitute one of the datasets studied: for each image and each pixel, the spectral study will be reduced to the study of these specific bands. Likewise the ratios : S / Fe, K / Fe, Mn / Fe, Cu / Fe, Zn / Fe are decisive for our study. These specific ratios will constitute another dataset studied. Study specific ratios is interesting to avoid some problems : as you write, the ink fades. As the ink fades, the amount of chemical elements also fades, however ratios between chemical elements remain unchanged.

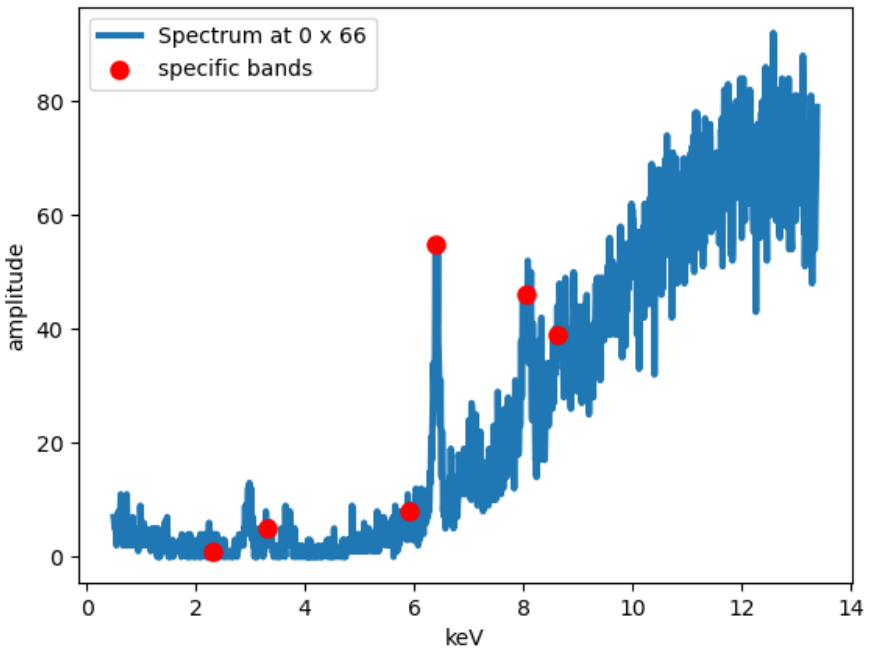


Figure 5 : specific bands (S : 2.313 KeV, K : 3.313 KeV, Mn : 5.9 KeV, Cu : 6.392 KeV, Fe : 8.042 KeV, Zn : 8.632 KeV)

In the field of Machine Learning, there are two main families of methods. There are supervised learning methods and there are unsupervised learning methods.

The first family requires pre-labeled training games: we use these games to train a model and test it, then we deploy this model to classify new individuals. The efficiency of such a model depends entirely on the training set (the more representative the set is, the better the prediction model will be, but care must be taken to avoid over-training). We can use a set of criteria (AUC score, Accuracy, ...) to evaluate the quality of a supervised learning model.

The second family does not require training and therefore does not need a pre-labeled data set. The efficiency of the methods of the second family is linked to the characteristics of the dataset to be processed (variance, covariance, correlation, presence of outliers, average, order of magnitude of the variables, ...). It is difficult to evaluate the quality of the algorithms of this family.

For our study, we do not have a pre-labeled set. Therefore, we need to focus on unsupervised learning methods or find a reliable way to label our data.

In the field of unsupervised learning we find the different families of clustering algorithms as well as the families of projection algorithms. There are several different clustering algorithms: K-means (see appendix II), hierarchical clustering (see appendix IV), density clustering (see appendix III), ... For the projection algorithms we find the dimension reduction algorithms such as Principal Component Analysis (PCA, see appendix V) and we find unmixing algorithms such as Independent Component Analysis (see appendix VI) or the non-negative matrix decomposition (NMF, see appendix VII) We will look at these different algorithms to try to solve our problem.

To complete our study, we can try some image processing methods or add some postprocessing : The writing is characterized by its form and the redundancy of its character.

Organization

For this internship, we work with agile methods. Favoring this type of operation allows us to adapt the work and to direct it according to the results obtained over short periods. Since we are trying to realize a proof of concept and the approach to follow is not defined beforehand, it is interesting to keep this agility.

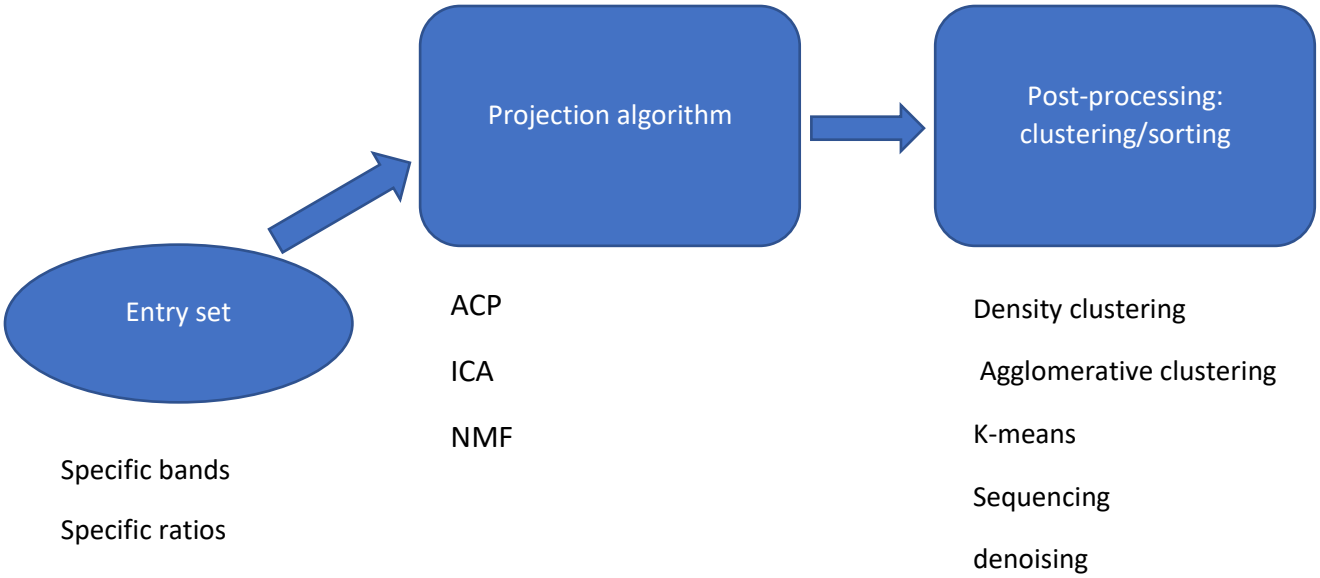
The results are communicated every week or two during video conference meetings. These meetings are also a way to define the way forward for the next sprint.

So far, four sprints have been completed :

- The first sprint, lasting two weeks, allowed us to take control of the data and to test various source separation strategies from machine learning and image processing.
- The second sprint, lasting one week, during which we sought to strengthen our understanding of certain Machine-Learning algorithms. During this second sprint we injected field expertise into our processing.
- The third sprint, lasting two weeks, during which we varied the parameters of our MachineLearning algorithms.
- The fourth sprint, ...

Strategy

The first strategy imagined can be summarized by the following diagram :



For this strategy the set of elements is flexible : one can work with different sets of inputs (consider specific bands or specific ratios for each redacted line), one can choose the projection algorithm and the post-processing to use. Since each line is written with a different ink (concentrations and chemical composition are variable from one line to another), the existence of a projection/post-processing combination offering a stable result quality is unlikely.

This strategy has a few negative points :

- Unless we can find a way to automatically determine the optimal projection/post-processing combination, we need to test all combinations for each line
- The parameterization of projection algorithms and post-processing is also expanded and increases the number of combinations to test

A second strategy that was devised was character recognition, using the tif images of our letters. This strategy has advantages :

- Inexpensive data acquisition : only photos of the letters
- Automatic processing common to the whole corpus

However, it also has disadvantages that affect its chances of success:

- Redaction ink would be interpreted as writing ink and would affect the results
- Requires an important labeling phase : cut the images by isolating each character and assigning a label manually

Results

Pre Study

For the study we focus on 3 images representing 3 cases of varying complexity. The first image is a favorable case, it is the first line of page 1 of the letter 1_1_8 :

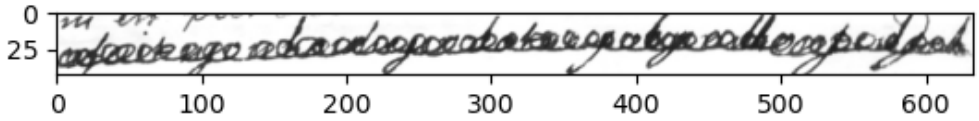


Figure 6 : line 1, page 1, letter 1_1_8

This case is favorable because we can easily separate the redaction ink from the writing ink due to different compositions (information highlighted by the REX II project). We can perform a pre-study on this first image in order to highlight certain characteristics and to estimate the chances of success of our algorithms. The first step is to visualize our variables, we choose to represent each specific band as a function of the Iron :

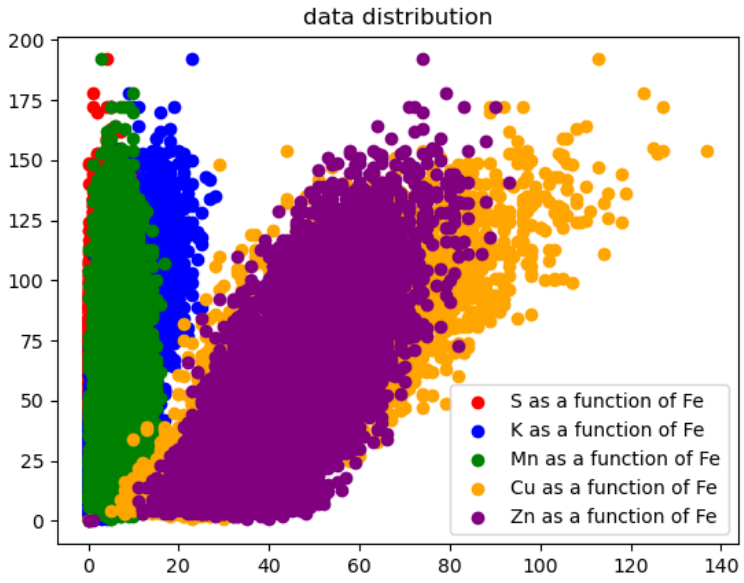


Figure 7 : pre study, picture 1 : data distribution

We can see that the distribution of our data is elliptical, moreover we can guess an important correlation between iron and copper as well as between zinc and iron which seem linearly dependent. It is difficult to interpret this type of representation in the sense that we do not clearly distinguish our individuals, however we have an idea of the distribution of the different variables and we do not manage to distinguish groups in these representations. We hypothesize that K-means clustering will have difficulty separating our inks since we cannot distinguish similarity groups. The REX I project has highlighted the discriminating power of the Cu/Fe ratio, we can therefore hypothesize that the Kmeans clustering will succeed in finding the underlying plot by focusing the study only on this ratio. The second step consists in studying the mean and the variance of the selected variables (specific bands and specific ratios) :

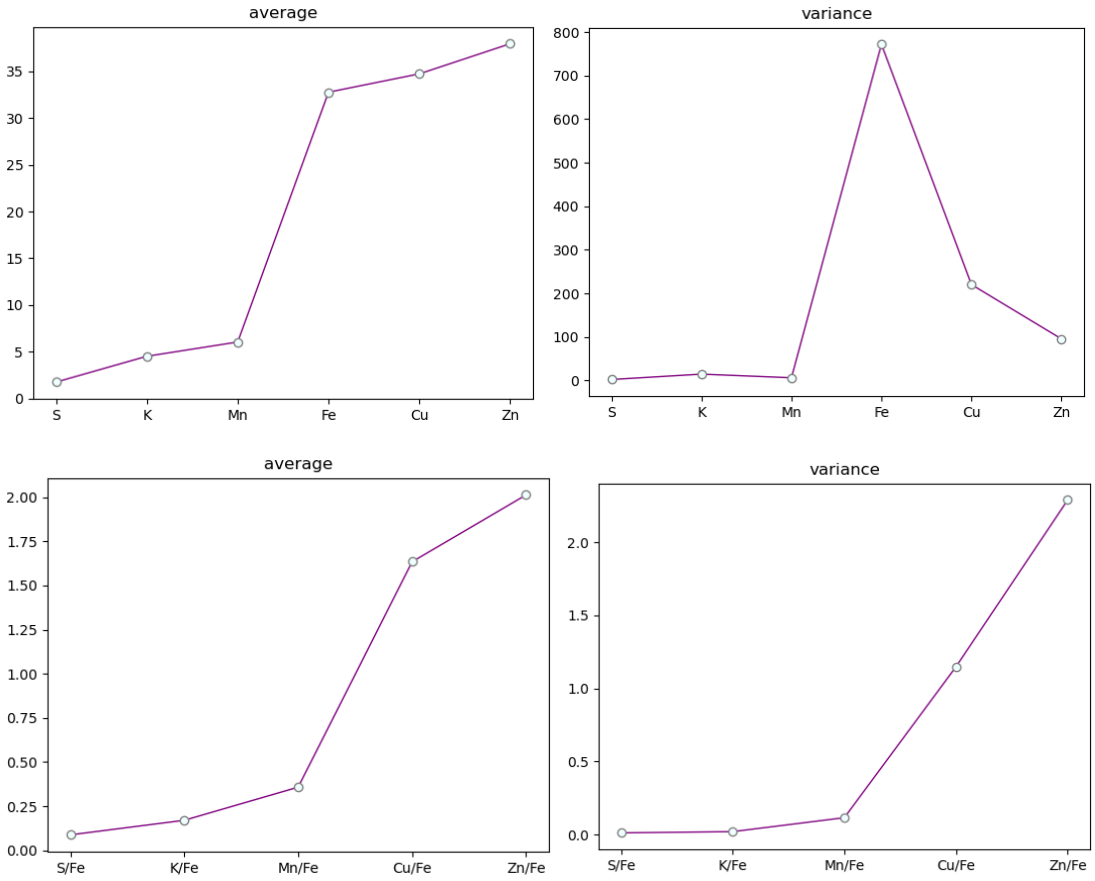


Figure 8 : pre study, picture 1 : average and variance

By studying the mean and the variance we get an idea of how the PCA works and of the interest of reducing our data (dividing each variable by its standard deviation, i.e. the square root of its variance). In the case of image 1, the metallurgical elements have high variances, but this variance becomes negligible when considering the specific ratios. The element with the highest variance is iron : iron is a primary constituent in Ferro-Gallic inks, so its variance is explained by the fact that the paper is not made of iron while the ink is mostly made of iron. Reducing our dataset does not seem necessary in this case, only iron has a significant variance, but iron is present in both the redaction ink and the writing ink: reducing our dataset will therefore have an impact on this set, however it will not have an impact on the results. We can determine the new origin calculated during the PCA whose coordinates correspond to the mean of each variable. The third step is to study the correlation matrices :

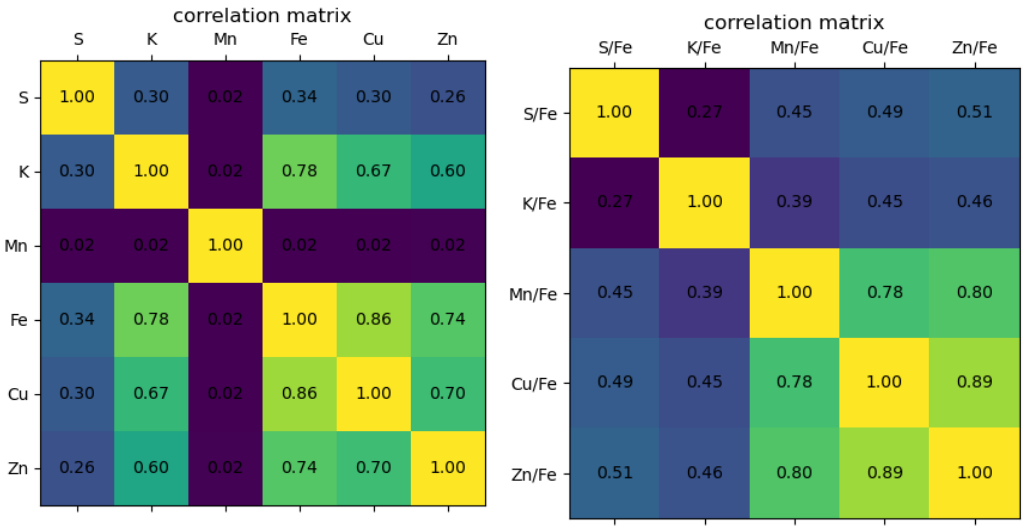


Figure 9 : pre study, picture 1 : correlation matrix

By observing the correlation matrices we can predict the behavior of the PCA : What is important is to observe sufficient diversity, we hope that the PCA will succeed in separating our sources and that each new axis calculated will carry the information related to our different sources (one axis for paper, one axis for redaction ink, one axis for writing ink). In this case, diversity seems to be conducive to the success of the PCA, and we will find the elements that are strongly correlated with each other carried on the same new axis. On the other hand, the elements that are weakly correlated to the others will be found to generate new axes of which they will be the main contributors. We can expect suitable results for this case.

This pre-study is carried out for a less favourable case, line 5 of page 6 of the letter 1_1_2 :

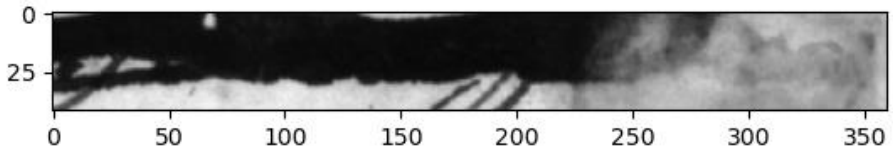


Figure 10 : line 5, page 6, letter 1_1_2

This case is unfavorable due to the ubiquity of the redaction ink and a greater similarity between the writing ink and the redaction ink. We display the distribution of our data :

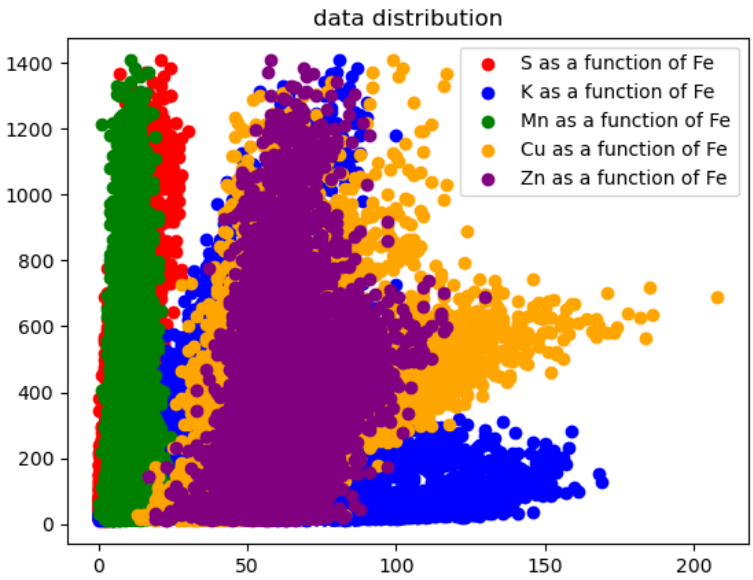


Figure 11 : pre study, picture 2 : data distribution

Again, we are unable to differentiate similarity groups in the distribution of our variables : we assume that K-means will have difficulty separating our inks. The correlations between copper and iron and between zinc and iron seem less important than in image 1. We display the variance and the mean :

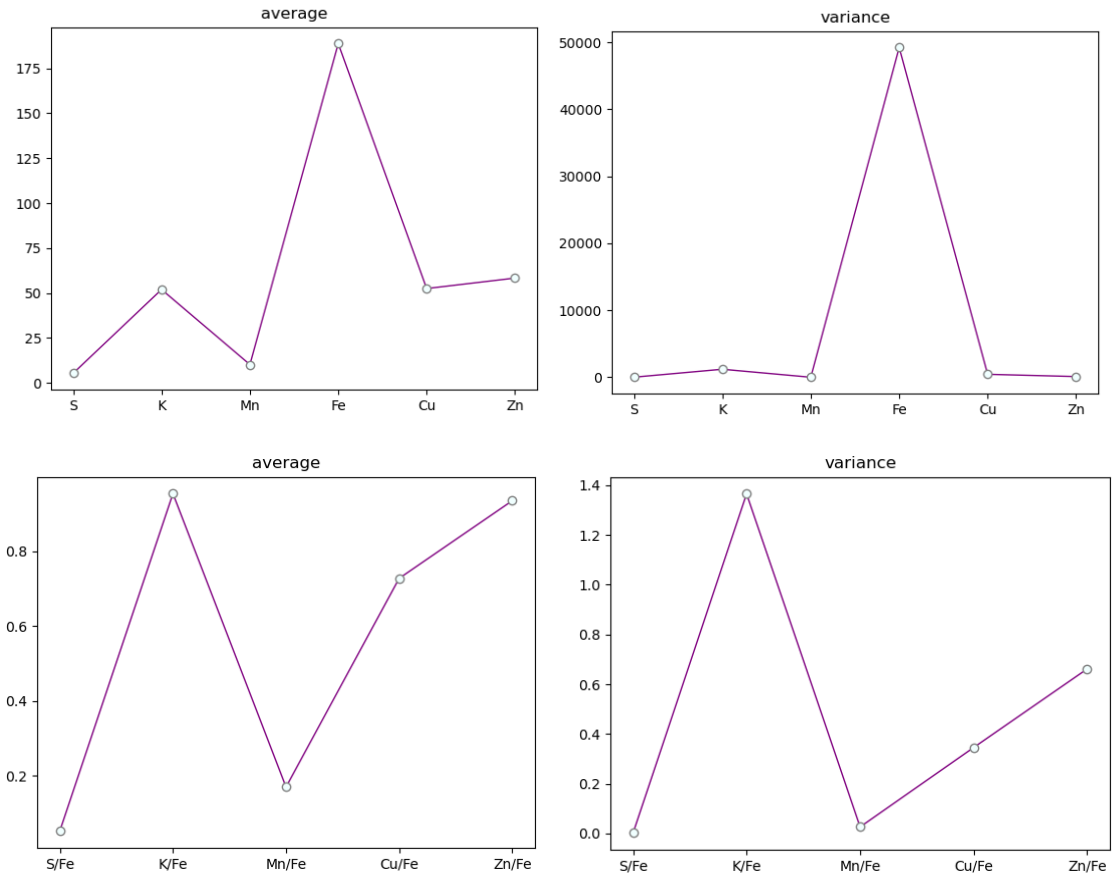


Figure 12 : pre study, picture 2 : average and variance

This time, only the variance of iron is a problem, however bleaching (the action of reducing our data) does not seem to be a determining factor for PCA : iron is a majority element in both the redaction ink and the writing ink, so its bleaching will not impact our results. We display the correlation matrices to complete the pre-study :

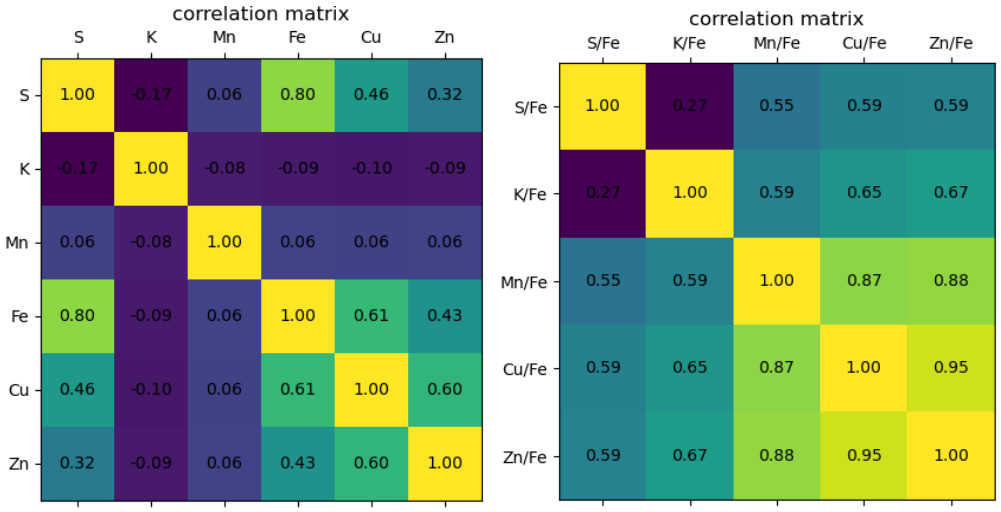


Figure 13 : pre study, picture 2 : correlation matrix

Compared to the first image, potassium no longer generates diversity (correlated identically with each element) and sulfur also generates less diversity : it is more strongly correlated with each variable. These differences reflect the use of different inks and/or an additional process (addition of metallurgical flakes to dry the ink). The chances of success of the PCA are reduced compared to the first image. Also the observations made on the distribution of our variables are confirmed: compared to the first image, iron is less correlated with zinc and copper.

The last case is the most unfavorable, it is line 1a of page 1 of the letter 1_1_2 :

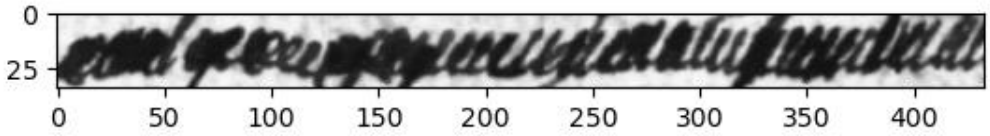


Figure 14 : line 1a, page 1, letter 1_1_2

This case is unfavorable due to the strong similarities between the inks of redaction and writing but also because of the influence of the back side on the front side. We trace the distribution of our data :

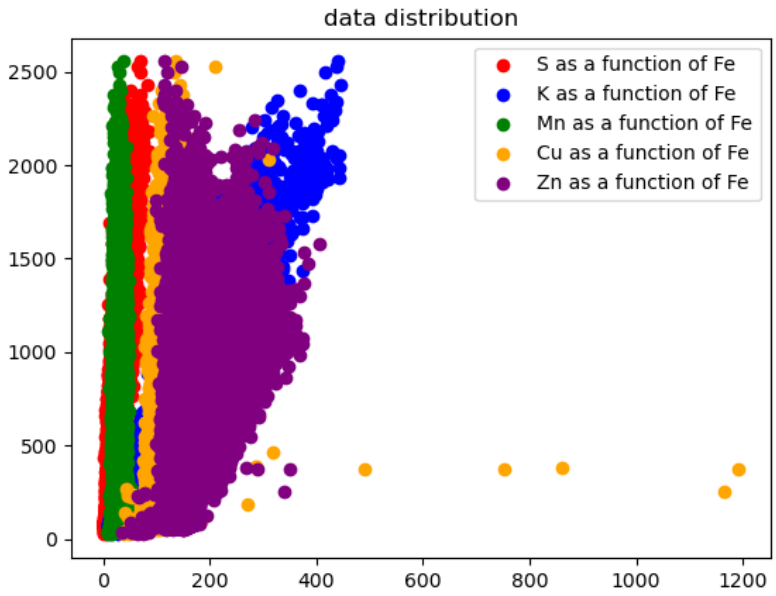


Figure 15 : pre study, picture 3 : data distribution

We observe outliers that disfavor the use of K-means. These outliers may be due to the influence of the back side on the front side. The presence of outliers makes the interpretation difficult.

Once again the study of the variance reveals that the variance of iron is particularly high, however the question of bleaching is of little interest for the same reasons mentioned in the pre-study of the previous images. We study the correlation matrices :

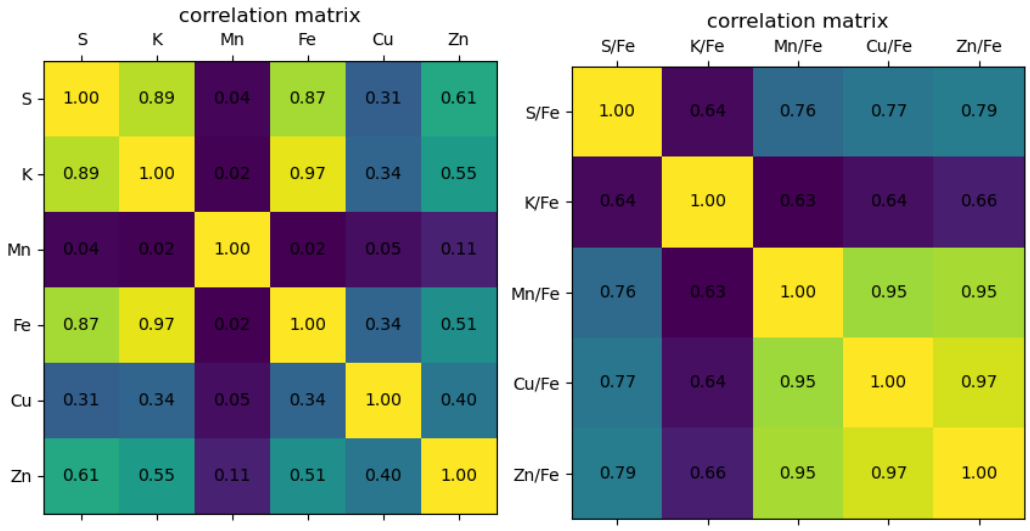


Figure 16 : pre study, picture 3 : correlation matrix

The correlation matrices show us that the correlations between the metallurgical elements have further decreased. As for the second image, compared to image 1, our diversity seems to be reduced and will influence the chances of success of the PCA.

Results

• K-means and initial centroids

During this study we will work by varying a number of parameters. Among these parameters, we find the choice of the initial centers for the clusters. We will work with a choice of centers called intelligent and proposed by the scikit-learn library ("k-means++" which chooses the best barycenters according to the inertia of the clustering : K-means++ tries to minimize the intra-cluster inertia. To achieve this, it repeats the initialization n times, n being a parameter that we can define). We will also work with an arbitrary choice of barycenters :

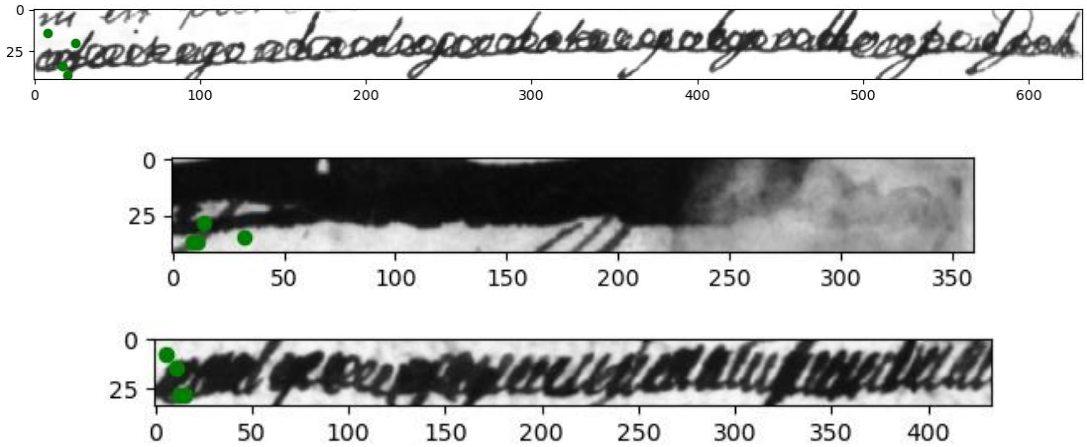


Figure 17 : arbitrary choice of barycenters for our three images (XRF data for green pixels)

We choose specific pixels as initial barycenters: we try to choose a pixel representing the paper of the letter (white area of our images), a pixel representing the writing ink (we choose a pixel that black that exceeds the redacted line : there is a good chance that it is a pixel belonging to the writing character that exceeds lines like the "f" or "p"), a pixel of redaction ink (we place ourselves on the line at a place where the density of ink is important) and a last pixel to represent the noise cluster (we try to put ourselves at the interface between the paper and the writing ink).

The other parameters that we will vary are :

- The number of clusters (3 : paper, writing ink and redaction ink or 4 : paper, writing ink, redaction ink and noise)
- The bands and ratios chosen (the clustering can be done on the whole set or on portions by choosing for example to take only the Cu/Fe ratio for all the pixels)

• K-means on specific bands (S, K, Mn, Cu, Fe, Zn)

For the first image, with the set of specific bands and a "k-means++" initialization, the K-means clustering gives the following results

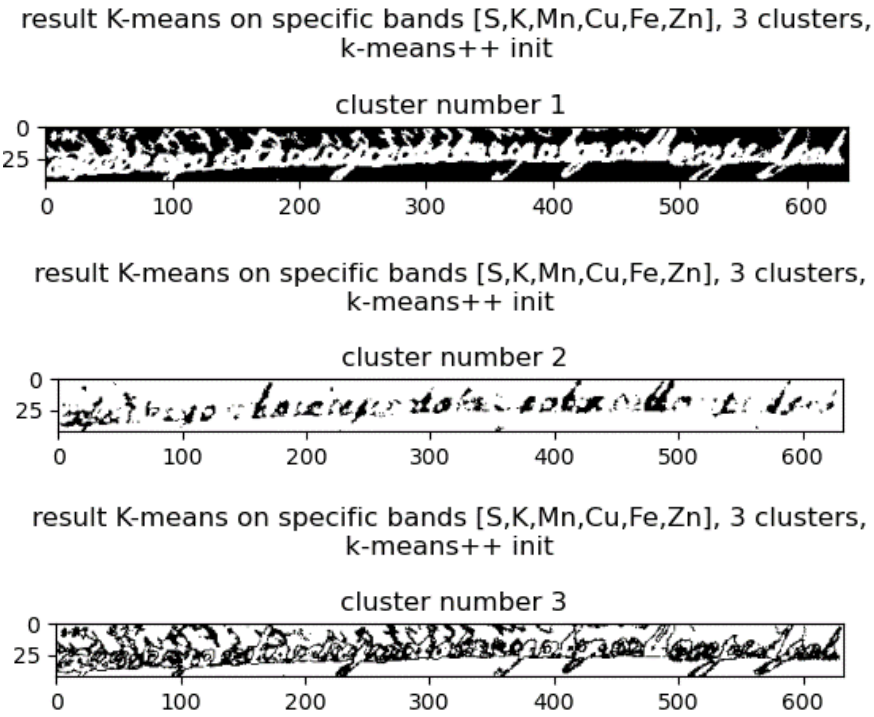


Figure 18 : clustering on all specific bands for 3 clusters and on image 1

Although the results are not bad, they are still insufficient and force us to find a complementary method to find the underlying plot. Moreover, these results are not replicable on more complex cases like image 2 :

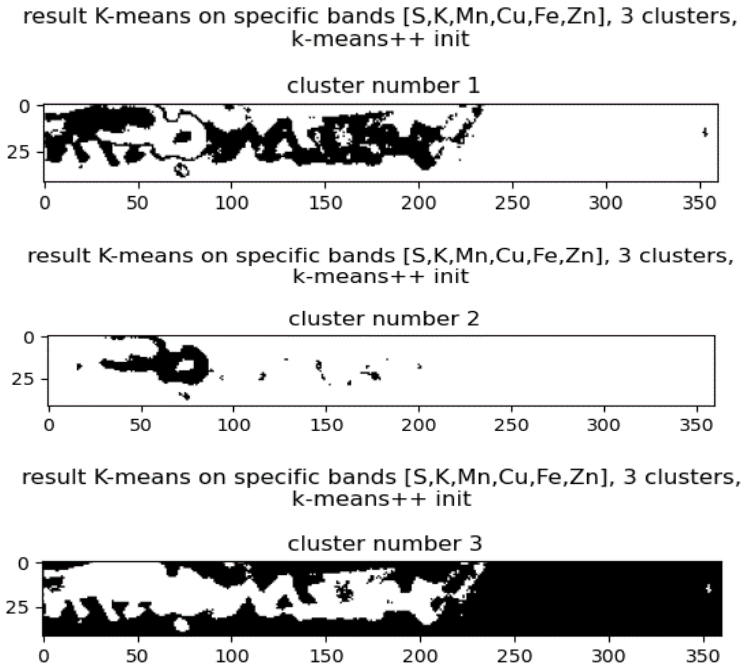


Figure 19 : clustering on all specific bands for 3 clusters and on image 2

Thus, even if we can get by with k-means for favorable cases by using the specific bands coupled with a complementary method, this same tool with the same parameterization for complex cases is insufficient.

The addition of a class to separate the noise from the rest of the clustering has little influence on our clustering :

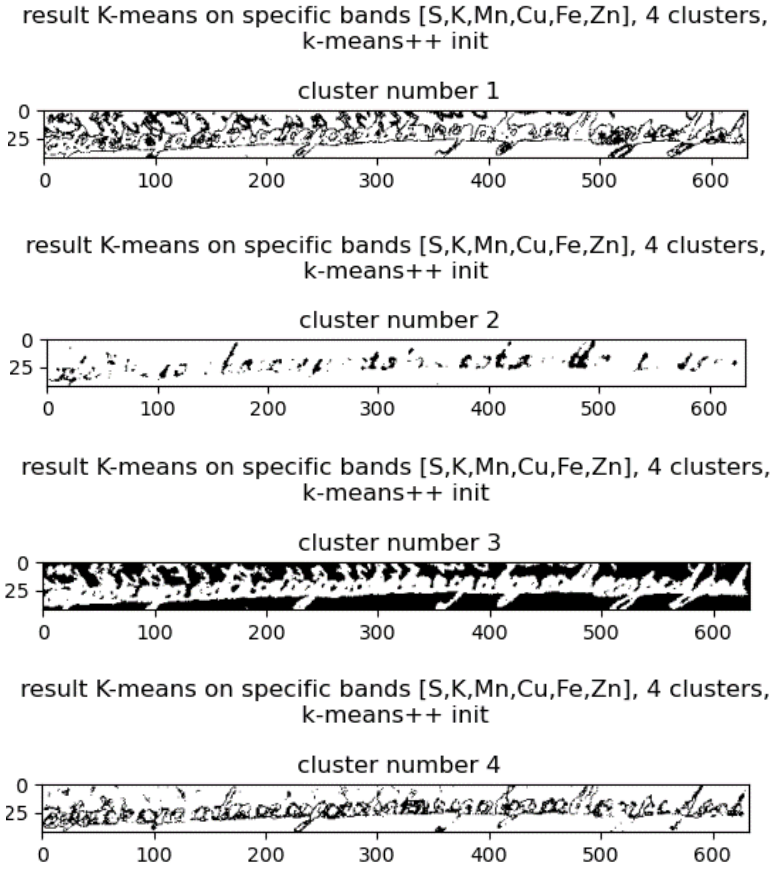


Figure 20 : clustering on all specific bands for 4 clusters and on image 1

Even if the results are not conclusive with the use of an additional cluster, this option remains interesting because it allows a better precision in certain cases :

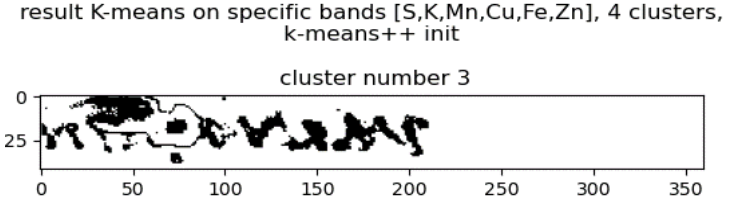


Figure 21 : clustering on all specific bands for 4 clusters and on image 2 (cluster 3)

On image 2, with all the specific bands, the addition of a fourth cluster allows a better separation of the inks. Generally speaking, the idea of a fourth cluster is interesting: the results produced are variable and can improve or reduce the quality of the results.

The choice of the initial barycenters did not impact our results, the clustering ends up converging towards the same result (the clusters are ranked: we are almost sure that the first cluster will be the one of the paper, the second the one of the writing ink, ...).

- K-means on specific ratios

Clustering with specific ratios is more interesting : we avoid the problems of ink intensity which runs out during the writing. Indeed, the ratios remain unchanged whatever the amount of ink deposited on the paper. Since K-means is sensitive to the shape of our data, using ratios is a good idea. So the clustering is of better quality with the specific reports.

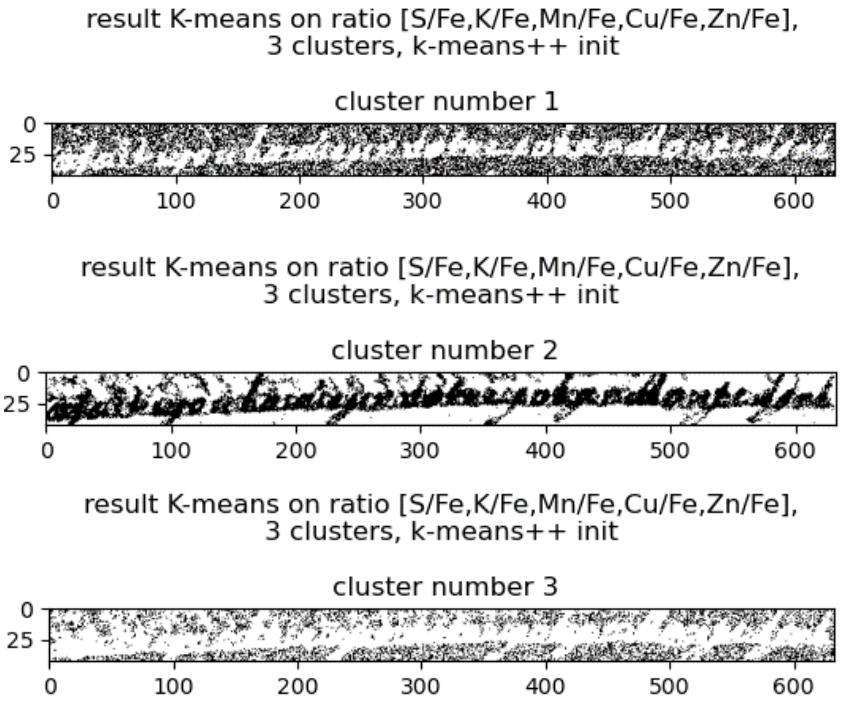


Figure 22 : clustering on all specific ratios for 4 clusters and on image 1

Moreover, in some cases (almost all the time), the addition of the fourth cluster offers very interesting results and by coupling this addition with the choice of subset with less unfavorable dynamics, we obtain suitable results :

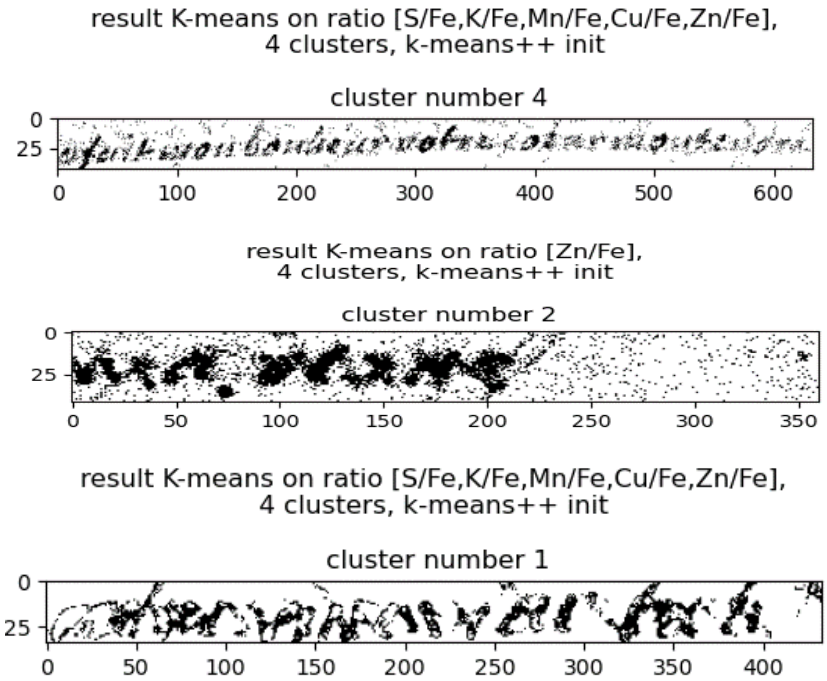


Figure 23 : suitable results according to the chosen subset and using 4 clusters for the 3 images (1 to 3 from the top)

As for the specific bands, the choice of the initial barycenters does not impact the clustering.

- K-means conclusion

K-means is a fast and easy to use clustering tool. For our study, the results produced with Kmeans, depend on the case treated, the parameterization and require more or less post-processing of variable complexity. The use of a data set with low dynamics is recommended to obtain the best possible clustering, therefore specific ratios are more interesting than specific bands. Also some subsets of ratios are more interesting than the whole set of ratios.

My personal conclusion is mixed concerning this tool, the diffuse and non-replicable character of the results as well as the necessity to find a complementary method for each case make the use of K-means not very easy for the separation of our sources.

- ACP and specific bands

For the specific bands we can vary the input set (choose all the specific bands or a part of them), however we must be careful to keep the maximum diversity: the PCA feeds on this diversity to define a new base.

The results with all the specific bands vary from case to case depending on the complexity

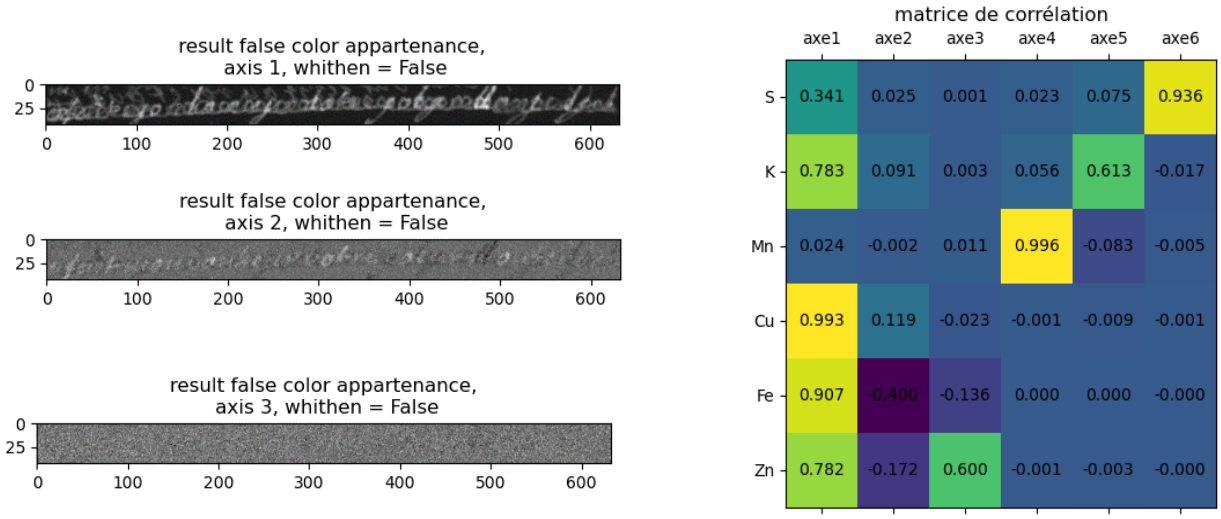


Figure 24: PCA result for all specific bands on image 1

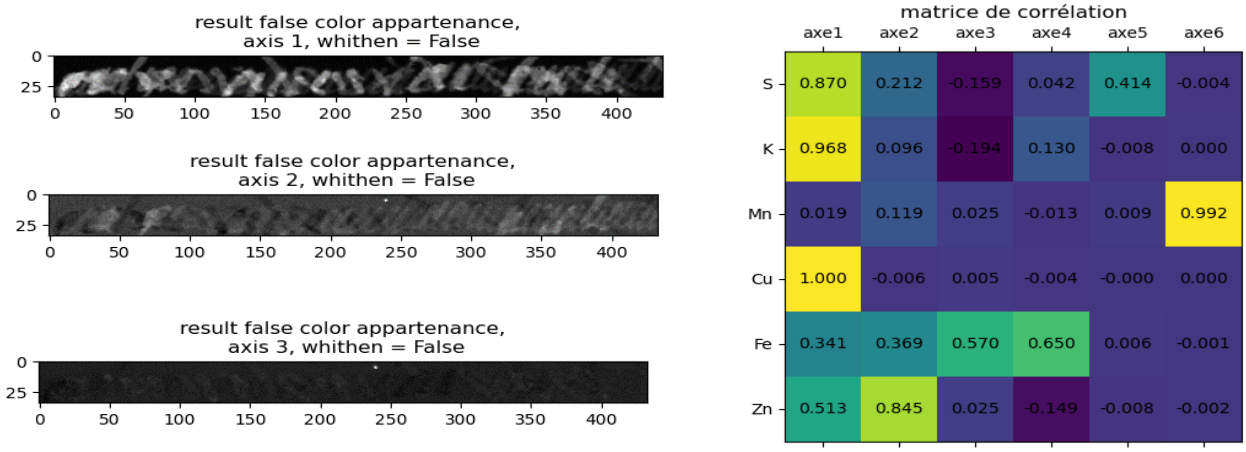


Figure 25 : PCA result for all specific bands on image 3

For the trivial case, PCA manages to find the underlying plot and with some post-processing, we would get slightly more accurate results than with K-means.

We can work with subsets, the first subset groups strongly correlated elements (S, K, Cu) and the second subset groups weakly correlated elements (Mn, Cu, Fe).

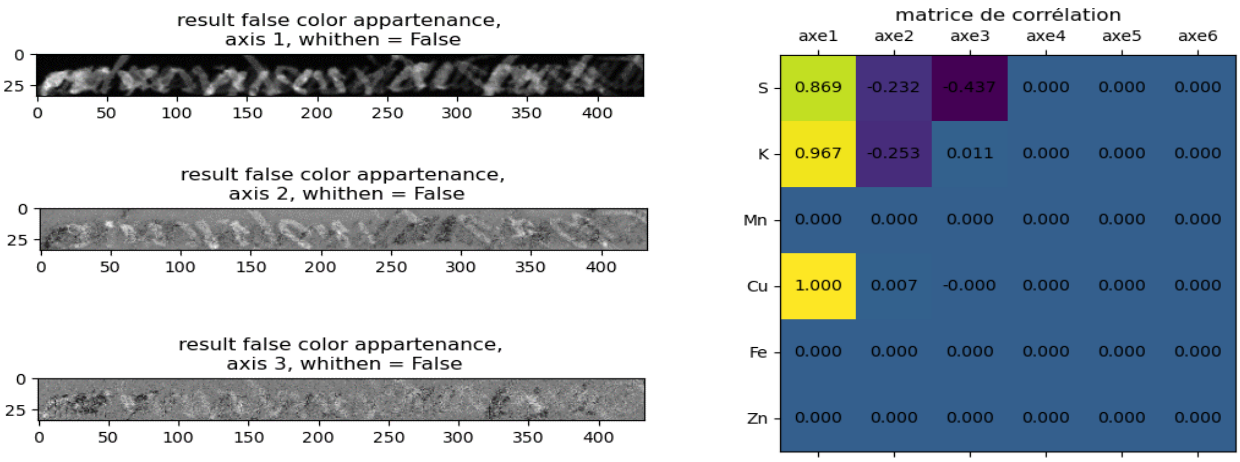


Figure 26 : PCA result for specific bands (S, K, Cu) on image 3

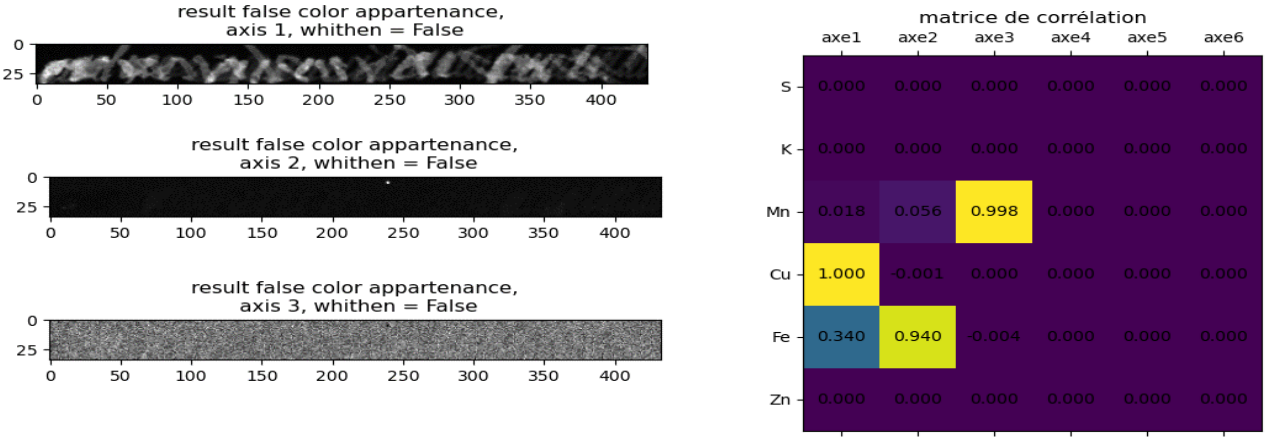


Figure 27: PCA result for specific bands (Mn, Cu, Fe) on image 3

The subsets do not bring significant changes. Also the bleaching does not bring any modification on the PCA and the output correlation matrix : we had foreseen this behavior. Only the iron was impacted by the bleaching but it is mostly present in the two inks to separate.

- ACP and specific ratios

For ratios, bleaching does not modify the output, the only parameter that can be varied is the input set. For K-means the use of ratios allowed a better precision because K-means is sensitive to the dynamics of the data set, however the PCA is built from this dynamic. Thus we can expect that the use of ratios does not bring any significant improvement for the PCA : the elliptical shapes will be softened but the correlations will be preserved from one element to another.

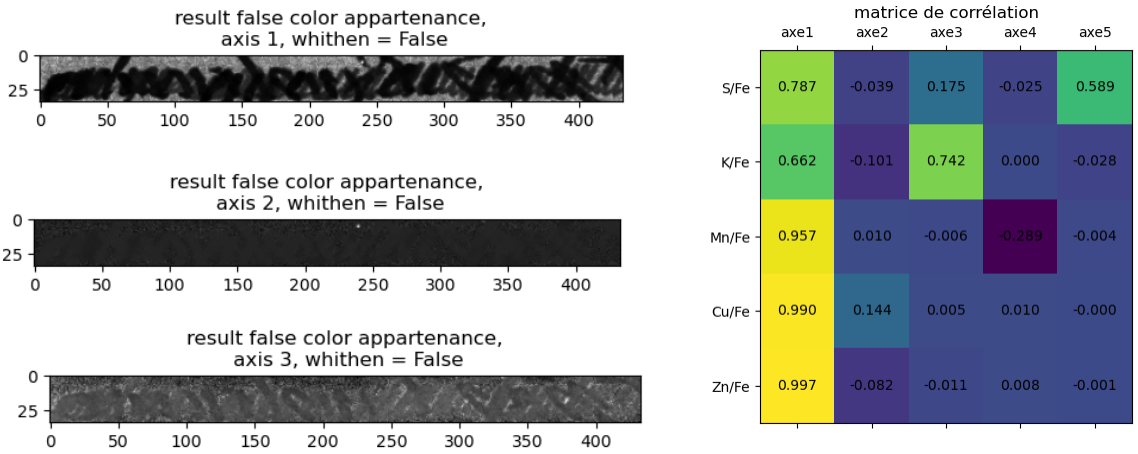


Figure 28 : PCA result for the set of specific reports on image 3

The results obtained with the ratios are similar to the results obtained with the specific bands. The use of ratios simply changed the dynamics of the data set. Yet the PCA was little influenced by this change : in reality the dynamics erased by the ratios is the variation of the amount of ink put on the paper, this problem had an influence on K-means but does not disturb the PCA at all.

As with the specific bands, the use of subsets does not produce a significant change in the result.

- ACP conclusion

The PCA is not intended to perform a classification, however we started with the idea that such an analysis could nevertheless allow us to separate our sources : ideally each source would have been carried by one of the new axes calculated by the PCA. For trivial cases this scenario seems feasible, we can see a part of the underlying plot and we can imagine post-processing to isolate this plot. However, for non-trivial cases, the PCA shows its limits and the underlying plot is not obvious to visualize.

PCA produces results close to those of K-means, and it is important to keep in mind that these are two algorithms of different families, performing two different operations. PCA reduces the dimension and projects on a base where each variable is decorrelated, K-means groups similar individuals.

- ICA and post-treatment

For ICA, we will try to set up post-processing. The results obtained are little affected by the parameterization: the main parameters influence the convergence speed. The parameters that could influence the final result are the choice of the number of sources to extract, the choice of the input set and the initialization matrix. Regarding the number of sources we chose a fixed number of sources (3) : paper, redaction ink and writing ink. For the input set, we will focus on the complete sets of specific tapes and specific reports. For the initialization, we choose the default initialization.

The objective is to see the contribution of post-processing coupled with ICA. The post-processings that we implement are :

- Thresholding on each ICA source (individuals above the threshold will receive the value 0 and individuals below will receive the value 1)
- Spatial de-noising. For each individual, we look at its closest neighbors (for each pixel we look at the 8 pixels surrounding it) and according to the number of neighbors being considered as ink, we decide that the individual is or is not ink (we assign it the value 1 or 0)
- We apply a K-means clustering on the source : the source or more precisely the result of the projection on the first axis calculated by the PCA is a result in gray scale. Applying K-means on such a result is like looking for the optimal thresholds to separate our individuals (we arbitrarily choose to make 3 clusters).

For all these post-processings, it may be interesting to study the influence of the parameters (choice of threshold, number of threshold neighbors, number of clusters, ...). In the future, we can look at other post-processing such as the use of agglomerative clustering (CAH) or density clustering (DBSCAN). We can also apply some of these post-processings on the previous results.

- ICA : specific bands and ratios

We start with a study on image 1 :

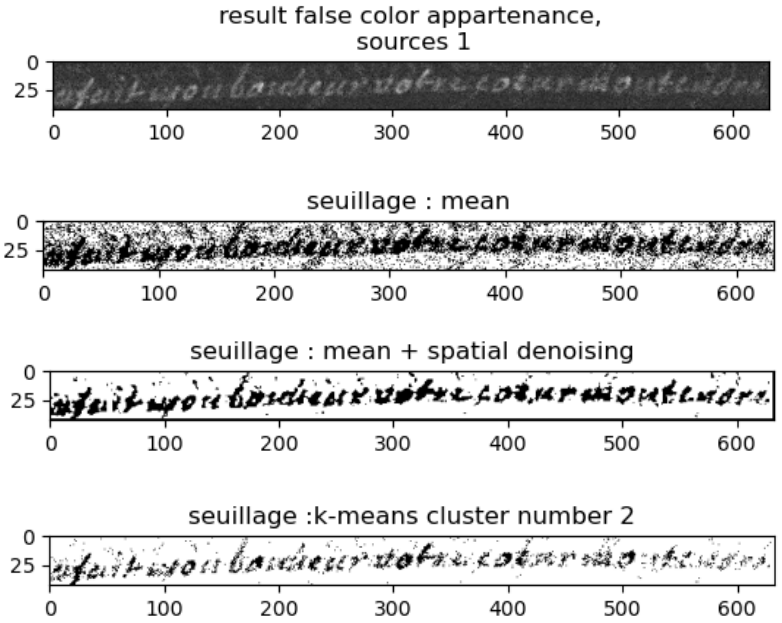


Figure 29 : ICA result followed or not by various post-processings: specific bands, image 1

We can also observe the results for specific ratios :

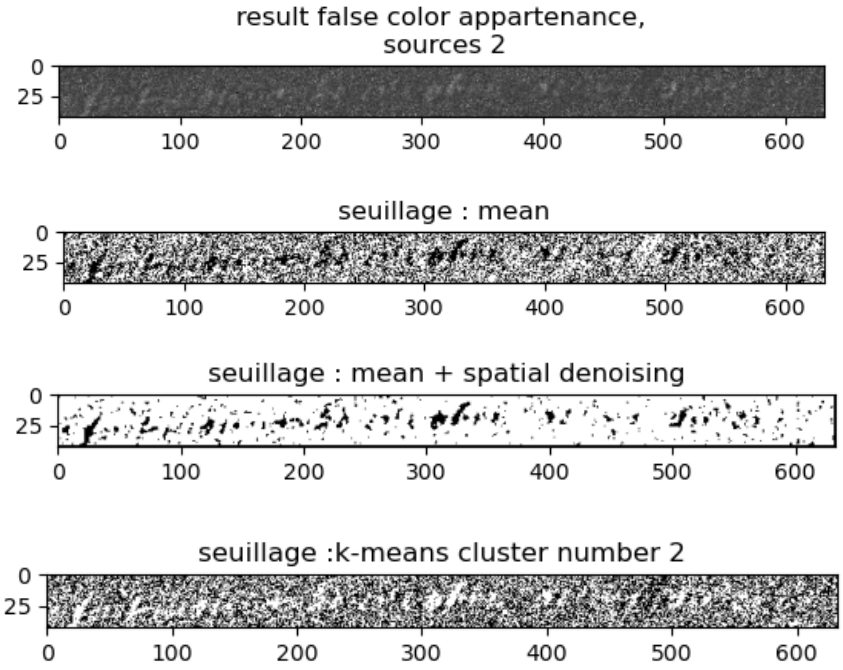


Figure 30 : ICA result followed or not by various post-processings: specific reports, image 1

For image 1, the specific ratios produce worse results than the specific bands. We can observe the results obtained for image 2 :

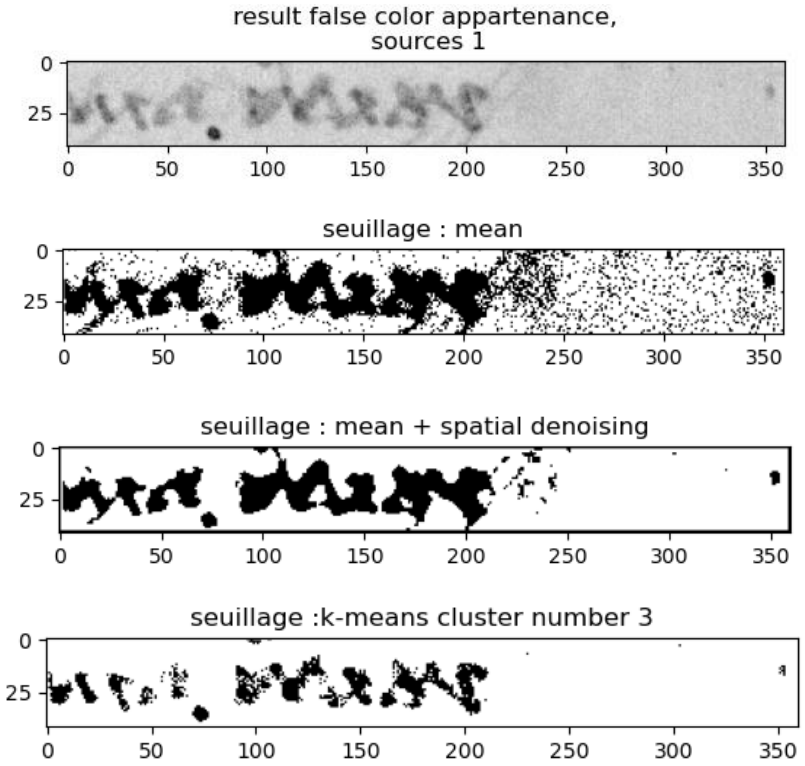


Figure 31 : ICA result followed or not by various post-processings: specific bands, image 2

We can easily distinguish an "a" and a "d".

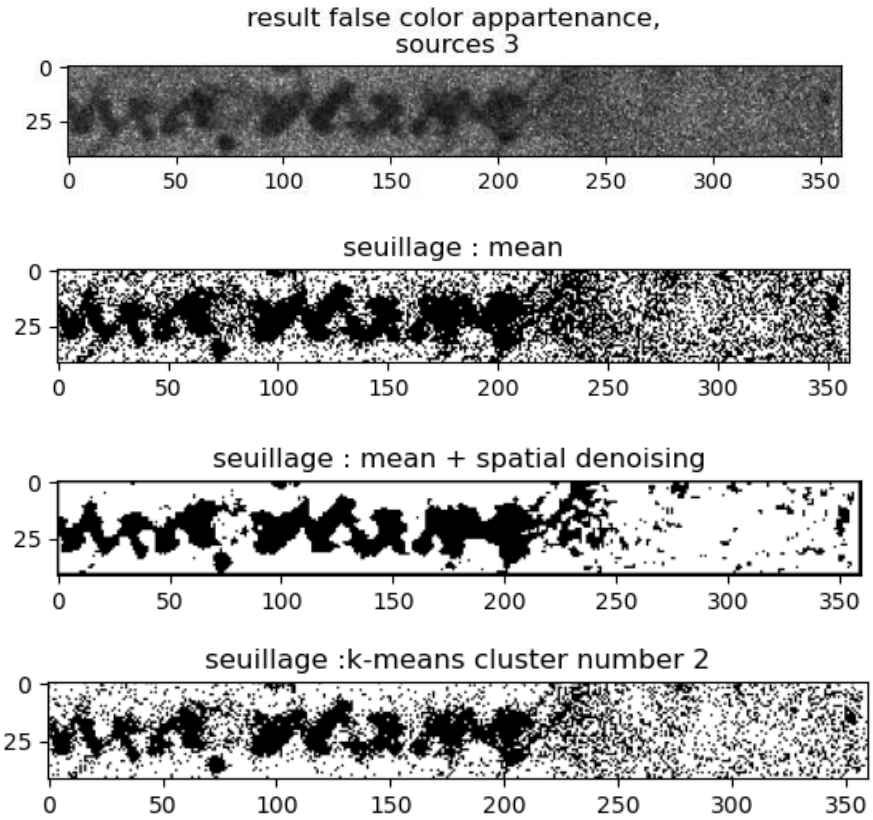
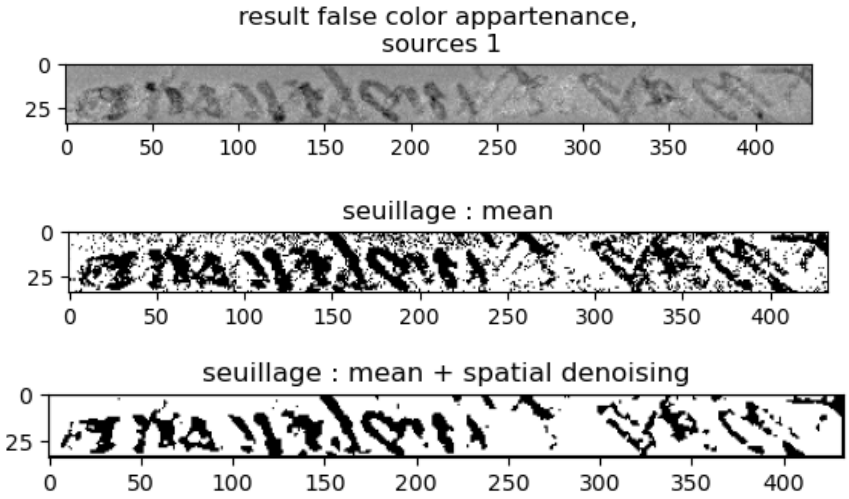


Figure 32 : ICA result followed or not by various post-processings: specific reports, image 2

For this case, no set seems to be necessary. Specific ratios and specific bands allow to recover similar plots.



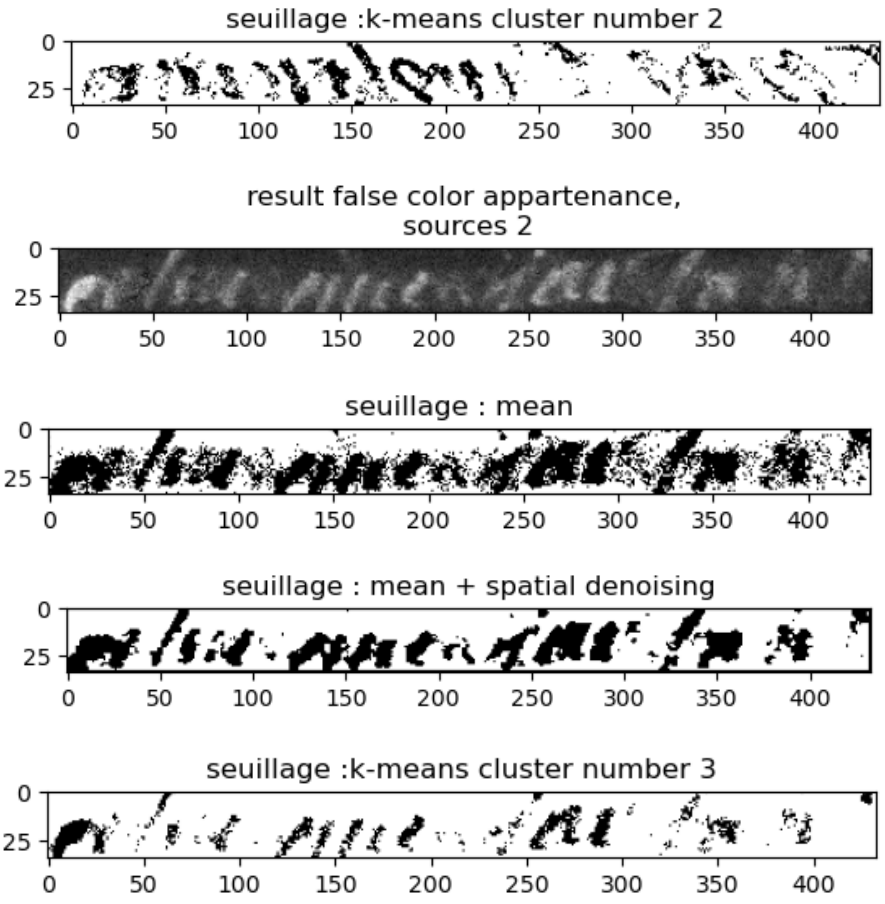


Figure 33 : ICA result followed or not by various post-processings: specific ratios, image 3

With this last result we have the impression that the ICA decomposition has separated the writing ink into two different sources. Our strategy may therefore fail in some cases.

- ICA strategy on the 1_1_8 letter

To test the effectiveness of the strategy, it is deployed on all the redacted lines of the letter 1_1_8. We therefore apply an independent component analysis and for each source we apply two post-processings :

- A threshold (average)
- A spatial de-noising

We recover the source that seems to correspond to the writing ink and we apply the two treatments.

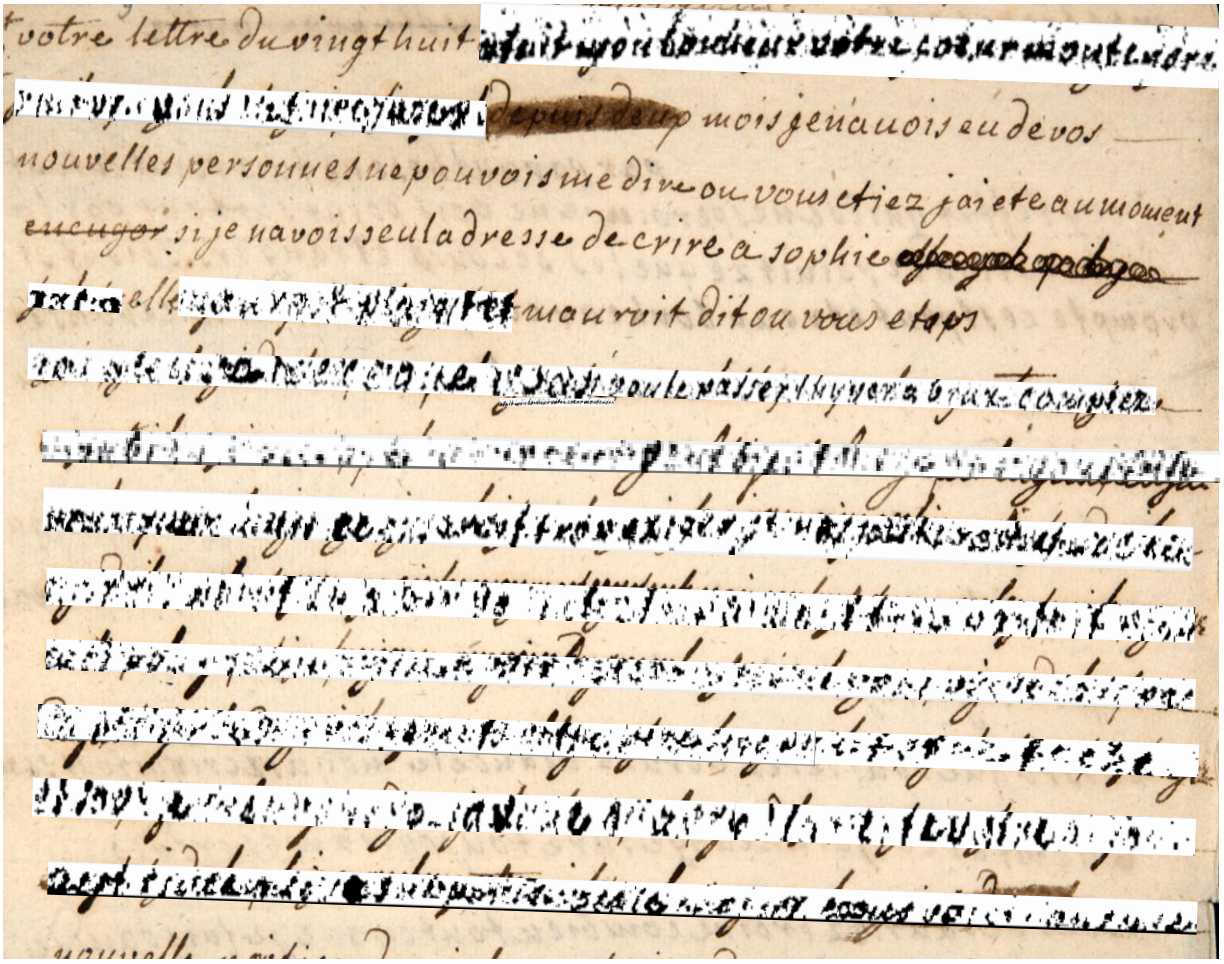


Figure 34 : ICA strategy + threshold + de-noising on the letter 1_1_8

The strategy does not succeed in finding the underlying plot for all lines. However we have very good results for some lines. The differences from one line to another are caused by several factors :

- ICA can separate the writing ink on two different sources
- CA can fail to find the writing ink (more precisely it can couple writing ink and noise on the same source)
- The thresholding may not be the most optimal (the mean may not be the right threshold to consider, one could consider a different interval or threshold)
- The spatial de-noising is too important (we can refine this de-noising, currently if at least 6 neighbors are considered as ink then the pixel is considered as ink too)
- The data set used is the set of specific bands for each pixel

We can play on all these factors and we can also change the components of the strategy (choose NMF or ACP instead of ICA or change the post-processing).

Conclusion

The results obtained so far are not conclusive. The tools used produce variable results whose quality cannot be replicated from one case to another. Moreover, the strategy imagined, which consists of coupling a projection method with one or more post-processing operations of varying nature, does not seem to be conclusive either : each line requires a different projection/post-processing combination.

For the following, a literature review is necessary to explore new avenues. Among the literature studied it may be interesting to study the criteria of distance between spectra. Also we had considered a second strategy which consist in character recognition : exploring this track can be interesting for our problem. We could try a character recognition on the raw images or on the images once we have "separated" the different inks: this recognition could also be done on the results of the REX II project.

This internship has contributed to reinforce my knowledge and deepen my knowledge in particular in the field of Machine Learning : The previous knowledge has been consolidated by the practical application and interpretation on concrete cases, as for the new knowledge they are the result of research to find new strategies to solve the problem. Even if the majority of the contributions of this internship concerns Machine Learning, this internship is a multidisciplinary internship and many tracks were explored allowing the contribution of new knowledge in other disciplines such as vision (use of convolution matrix, research of spectral distance criterion for images, ...), chemistry (understanding of the composition of Ferro-Gallic inks, understanding of the popularity of these inks, ...) and physics (X-ray fluorescence spectroscopy).

In addition to the technical contributions, this internship has contributed to improve my work methodology and has allowed me to understand the expectations in a multidisciplinary project : conducting a multidisciplinary project requires that each party can understand a minimum of the whole project, for my part I must understand the work done during the REX and REX II projects. In addition to understanding, we need to be able to communicate our results clearly so that they are understandable to all. Working on a multidisciplinary project requires a certain flexibility. This internship allowed me to become aware of this aspect.

Annexe I : X-Ray fluorescence spectroscopy

There are several different spectroscopies. The best known are the spectroscopies in the visible range. We can cite the emission and absorption spectra related to each chemical element or the hyperspectral images allowing the acquisition of the entire reflectance spectrum of an image in the visible range.

In our case the X-ray fluorescence spectroscopy can be related to an emission spectrum in the X-ray domain. The study of this spectrum allows us to identify the chemical elements present in the image studied.

The image studied is bombarded with X-rays and the materials in the image re-emit part of these rays : this is called X-ray fluorescence. The spectrum of X-rays emitted by the material is characteristic of the composition of the sample, by analyzing this spectrum, we can deduce the elemental composition, ie the mass concentrations of elements.

The spectrum can be studied from the energy or from the wavelengths. Thus we observe the mass concentrations of our elements from the energy bands (KeV) or from the metric bands. Each band is associated with a chemical element. We find for example :

- The Kα band of sulfur at 2.313 KeV
- The Kα band of potassium at 3.313 KeV
- The Kα band of manganese at 5.9 KeV
- The Kα band of iron at 6.392 KeV
- The Kα band of copper at 8.042 KeV
- The Kα band of zinc at 8.632 KeV

There are several bands characterizing the presence of different chemical elements. In our study the most accessible and defining bands are the previously mentioned Kα bands.

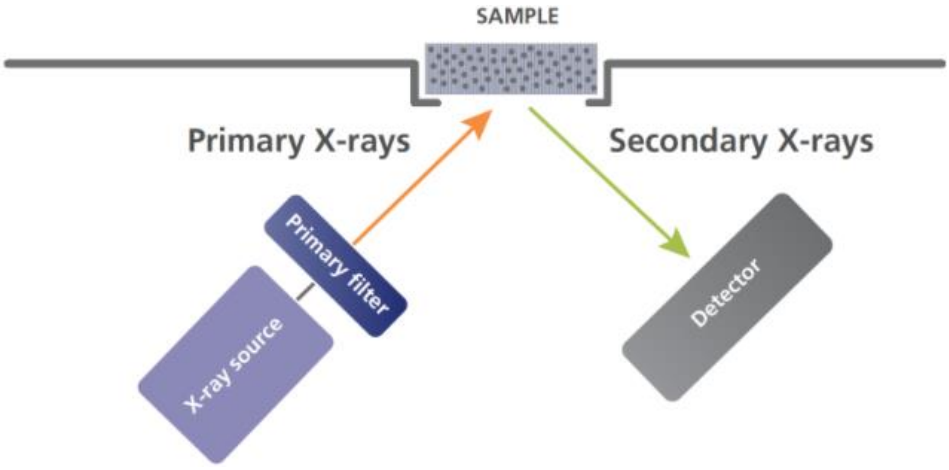


Figure 35 : fluorescence X-ray principle

Filter is used to select specifics X-ray.

Annexe II : How K-means clustering works

The K-means algorithm is an iterative algorithm whose goal is to classify individuals in a dataset into groups. This classification is more commonly called clustering and uses the variables of the dataset to form similarity groups. These groups are more commonly called clusters and each of these clusters groups a set of similar individuals.

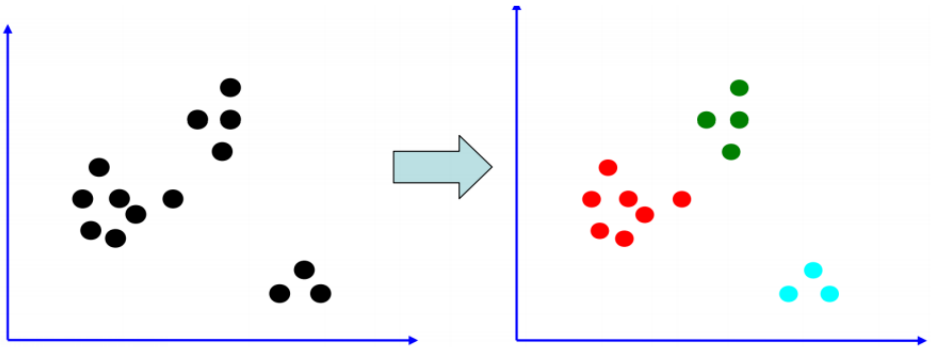


Figure 36 : k-means clustering (image taken from P. CARRE's data analysis course)

To realize this distribution in clusters the algorithm uses an iterative method: at each iteration the center of the clusters is recalculated and then we identify the individuals who belong to the different clusters. To define the members of a cluster we use the distance with the cluster centers : an individual belongs to the cluster whose center is the closest. The stopping condition of this algorithm is the convergence. After defining a tolerance threshold for convergence (ε), we run the algorithm and it stops once the distance between cluster centers from one iteration to the next is less than or equal to ε.

Thus the K-means algorithm works by recalculating the cluster centers at each iteration. Rather than using a random initialization, it can be important to fix the initial centers with additional information. The initialization is crucial and has an impact on the final result.

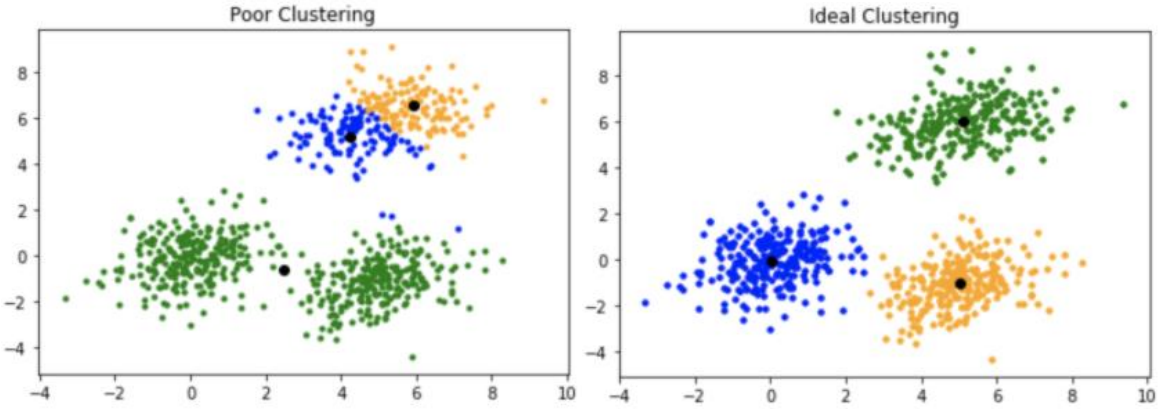


Figure 37 : initial centroids affecting k-means results

This algorithm has many advantages :

- Relatively simple to implement.
- Scales to large data sets.
- Guarantees convergence.

But it also has its disadvantages :

- Choosing number of clusters manually (can generate the wrong clustering)
- Dependent on initial centroids
- affected by outliers (outliers is included in clustering and can disturbed the centroids calculation)
- troubled by the data dynamic (anisotropicly data group, unequal variance, unevenly sized data group)

For our study the K-means algorithm will not be sufficient, the dynamics of our datasets, the presence of outliers and the strong similarity of our redaction and writing inks are factors that influence the quality of K-means. However, we will be able to use this tool as a post-processing tool (it could be useful to perform a thresholding without having to search for the optimal thresholds).

Annexe III : How density clustering works

The operation of density clustering like DBSCAN is based on the direct neighborhood of individuals. The principle is simple : if we find enough similar individuals in a given radius then we all belong to the same group. This type of clustering works from end to end : for each individual we test the neighborhood and we group the similar individuals.

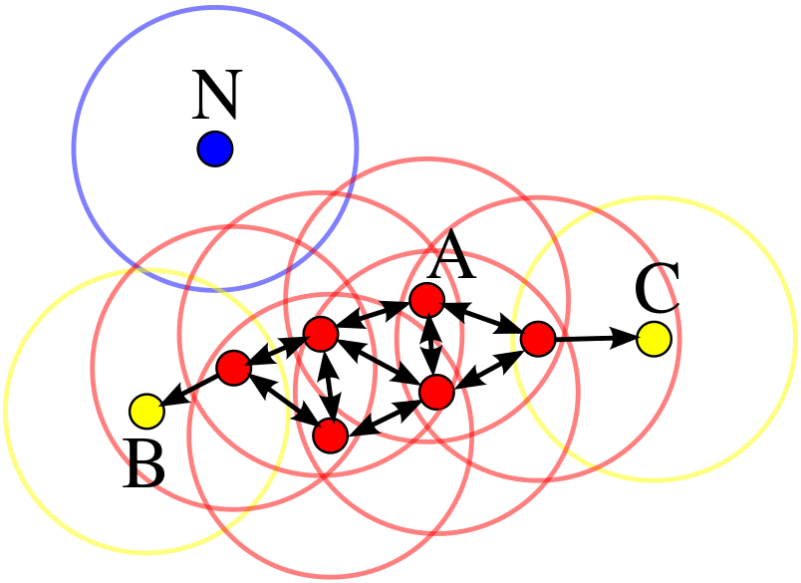


Figure 38 : DBSCAN principle

In this example the red points constitute the same cluster, the yellow points constitute a second cluster and the blue point is an outlier. For each red point we observe the number of neighbors in a fixed radius : here the rule is that if at least three neighbors are present in the radius then the studied point belongs to the red cluster. Therefore the young points constitute a second cluster because they have only one neighbor in their radius, while the blue point is an outlier because there is no neighbor in its radius.

In our study this tool can be interesting in post-processing to filter outliers. Moreover, the writing ink adapts to this behavior: the characters follow each other and are close to each other, so the notion of neighborhood is interesting to study.

Annexe IV : How agglomerative clustering works

Hierarchical ascending clustering (HAC) is one of the clustering by agglomeration. This clustering is iterative: at each iteration we redefine new clusters. In the case of HAC, we start by considering all the individuals as singletons. Over the iterations, we group the different singletons and the closest clusters two by two.

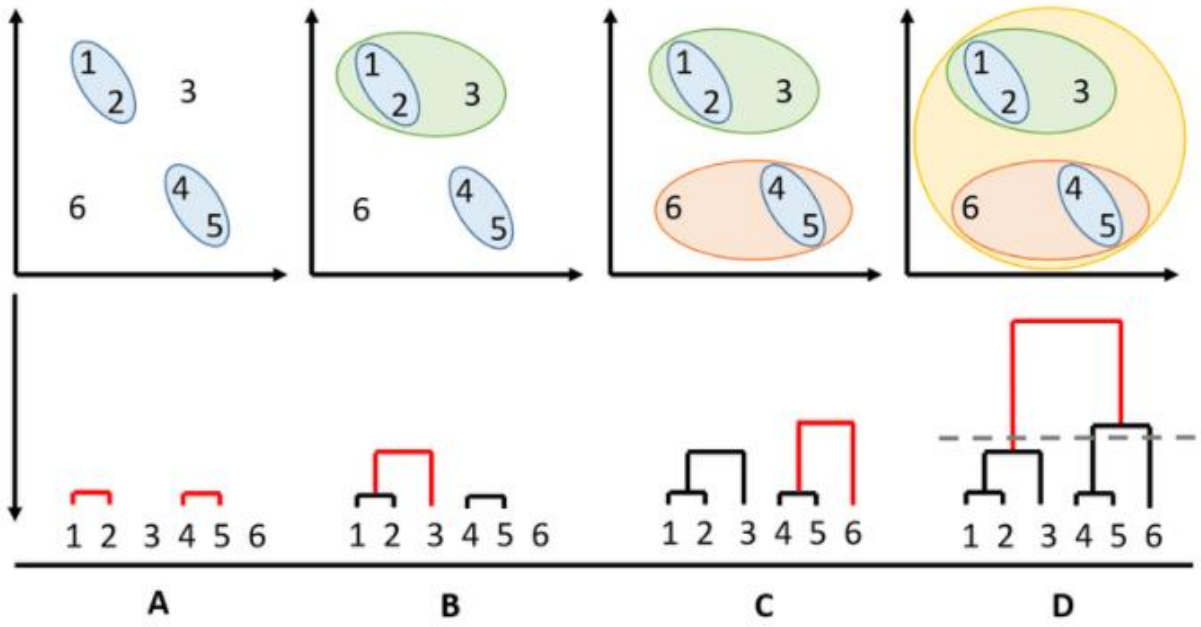


Figure 39 : how hierarchical ascending clustering works

A, B, C and D represent the different iterations of our algorithm: at each new iteration we compute a distance matrix between each group (cluster or singletons) and then we join the two closest groups. The distance criteria are multiple and depend on our application.

In our case this type of algorithm cannot be applied to all our data due to the computational power required: we must reduce the number of variables studied.

Annexe V : How Principal Component Analysis works

PCA or principal component analysis is a dimension reduction algorithm. The objective of this algorithm is to reduce the number of variables in a data set. To achieve this, the algorithm is interested in the dynamics of the data set. By observing the dynamics of the dataset, the algorithm will project all the individuals in a new base. This new base will have for origin the average of the initial variables (the coordinates of the origin of the new base will be the average of each variable). The axes of the new base will be such that they will preserve the dynamics of the scatterplot: the number of axes for the new base will be less than or equal to the number of initial variables. Each new axis is constructed by linear combination of the initial variables : if two initial variables are strongly correlated, then there is a good chance that they will be carried by the same axis.

The algorithm consists of several steps :

- We center our dataset: this means subtracting from each variable its mean (for all the individuals). By performing this step, we move our dataset around the origin of our new database.
- The second step is optional, it is the reduction of our dataset: it amounts to dividing each variable by its standard deviation. By reducing, we desensitize the order of magnitude of our variables while preserving the dynamics of the data set. Such an operation is similar to a change of unit for a standardized unit (standard deviation and variance = 1 for each variable).
- The third step is to determine the variance/covariance matrix of our centered (reduced) data set. To do this we start to define a diagonal weight matrix, each coefficient represents the weight of a variable (in our case we give the same weight to each of our variables or 1/number of variables). Once the weight matrix is defined then we can determine the variance/covariance matrix (data set * weight matrix * transpose of the data set).
- The fourth step is to define the new axes for our new basis : we diagonalize our variance/covariance matrix. The eigenvectors are the director vectors of our new axes. We have to be careful to make these vectors unitary and to arrange them in order of importance (we arrange them so that the diagonal matrix of the eigenvalues is in decreasing order). Thus the first axis will represent the largest share of the initial set and the following axes will represent a smaller share (the share represented is also decreasing).
- The last step is to project our game into the new database. The variables of the new database are decorrelated

The parameters we can influence are the starting set and whether or not to reduce the variables in this set.

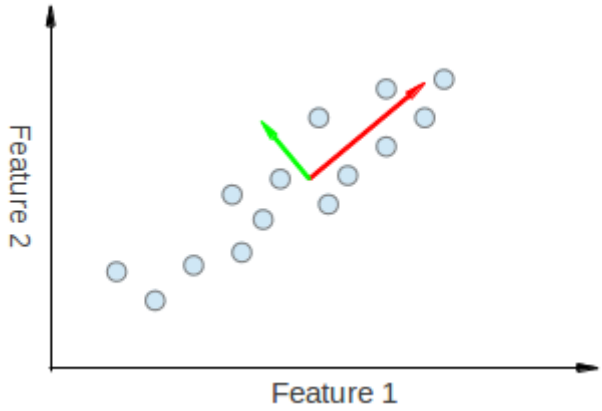


Figure 40 : ACP example (image taken from OpenCV web documentation)

In the example the initial variables are 2 : Feature 1 and Feature 2. The new base is composed of the red and green axes, with the average of Feature 1 and Feature 2 as origin. Since the red axis represents the major part of the dynamics of the initial game, then we can keep this single axis as the new base (we always try to keep at least 95% of the initial information and this part is reached with the first axes of the new base).

In our study PCA can be used to extract features: we will look for the underlying plots to be plotted on one of the new axes. If the similarities between the writing ink and the redaction ink are too great, then PCA will not be able to extract our underlying plots : PCA needs diversity because it bases its calculation on the conservation of this diversity and the factorization of correlations.

Annexe VI : How Independent Component Analysis works

Independent Component Analysis (ICA), also known as a solution to the blind source separation problem, is a basic change algorithm like PCA. However ICA is different from PCA. Indeed, where PCA tries to preserve the dynamics of the dataset by performing "factorization" (PCA finds the best orthogonal basis that decorrelates our data while preserving as much of the initial information as possible), ICA tries to find a new basis that decorrelates our data and where each axis is independent of the others.

When using this algorithm, we impose a number of axes/sources to be found, each individual is then considered as being a linear combination of these sources. The algorithm, by imposing the independence of the sources, ends up finding the combination for each individual.

To better understand this algorithm, we can illustrate it with the "cocktail party" problem: in a cocktail party with a large number of guests, there are many microphones. The objective is to separate each recorded voice by erasing all the others considered as parasites. In this example, each voice is a source and each microphone is an individual who has recorded a linear combination of each voice.

The statistical independence of two events is based on the fact that their union is empty :

$$P(A \cap B) = P(A) \times P(B)$$

In the case of the ICA, this amounts to having non-Gaussian distributions for our sources. Thus the knowledge of a source does not allow to predict the others. There are many ways to ensure this statistical independence, we can for example use the maximization of the neguentropy. We wish to maximize the neguentropy which means to get as far as possible from a Gaussian distribution.

From a mathematical point of view this criterion of neguentropy of a variable is equivalent to a difference between the entropy of a Gaussian distribution and the entropy of the distribution of the variable (the Gaussian distribution must have the same mean and the same variance as the distribution of the variable).

$$néguentropie\ (x) = entropie\ différentielle\ (G) - entropie\ différentielle\ (P)$$

$$néguentropie\ (x) = -\int G(x)\log\big(G(x)\big)\,dx - \left(-\int P(x)\log\big(P(x)\big)\,dx\right)$$

With x our independent variable, G the Gaussian distribution equivalent to P (same mean and same variance) and P the distribution of x. This calculation is iterative: at each iteration x and its distribution are recalculated in such a way that the distribution P moves further and further away from a Gaussian distribution.

Annexe VII : How Non-negative Matrix Factorization works

The non-negative matrix factorization is a data projection algorithm. We decompose our data into a factorization of two positive coefficient matrices.

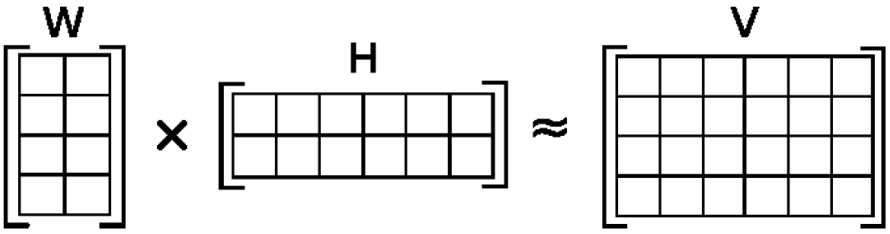


Illustration of approximate non-negative matrix factorization: the matrix V is represented by the two smaller matrices W and H, which, when multiplied, approximately reconstruct V.

Figure 41 : non négative matrix factorisation

To achieve this decomposition, the nmf attempts to minimize the following function :

$$0{,}5 \times \|V - WH\|_{loss}^2 + \alpha \times l1_{ratio} \times \|vec(W)\|_1$$
$$+\alpha \times l1_{ratio} \times \|vec(H)\|_1$$
$$+0{,}5 \times \alpha \times (1 - l1_{ratio}) \times \|W\|_{Fro}^2$$
$$+ 0{,}5 \times \alpha \times (1 - l1_{ratio}) \times \|H\|_{Fro}^2$$

Where

$\|A\|_{Fro}^2 = \sum_{i,j} A_{ij}^2$ represent the frobenius norm

$\|vec(A)\|_1 = \sum_{i,j} abs(A_{ij})$ represent an specific norm named L1 norm

$\|V - WH\|_{loss}^2$ represent the Frobenius norm or another beta divergence as Kullback-Leiber divergence.

At each iteration we redefine the matrices W and H taking into account the previous iteration until the convergence criterion is reached (minimum of the function reached).

In our study, the NMF is an algorithm that is placed between PCA and ICA : in some cases, the calculation of the NMF amounts to performing a PCA or ICA.

Ici un résumé